GE VERNOVA

Program Manager

2.4.1

User Guide

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Change History

Date	Change Description
December 18,2025	Program Manager 2.4: • Program Capacity Visualization • High-Level Metrics Overview in Dashboard • Curve Management
June 20,2025	Program Manager 2.3: • Update Program Management
June 15,2024	Program Manager 2.1: • Added User Interface Overview • Added VPP feature • Updated 2.1 UI updates
November 1, 2023	New document

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1. Solution Overview

1.1. Introduction

Program Manager is a module of DERMS acting as a comprehensive system of record for managing program, enrollment, and contract information in the utility's commercial context.

Program Manager enables utilities to implement methods for controlling DER behaviour and leveraging third-party or aggregator-managed DERs to provide grid support services. It allows a program administrator to define program rules and DER eligibility criteria to ensure enrolled DERs are dispatched according to program constraints. For example: programs could be created to manage flexible interconnections and capacity deferral programs, which increases grid equipment utilization and renewable hosting capacity.

Program Manager provides the following key business capabilities:

- Program creation and program management
- DER eligibility and enrollment
- Service provider management
- Event tracking and reporting

The Program Manager module can be deployed as a standalone application or as a modular add-on to the GridOS DERMS platform.

Programs are created to procure flexibility, energy, or other services from a service provider. A service provider can be an aggregator, a DER owner, a large industry, or residential customers, and are owners of DER assets.

Programs can be either passive or active. Passive programs provide incentives to modify DER behavior over long time periods, but they don't require monitoring or dispatch. Active programs dispatch specific DER or provide incentives or penalties to enrolled DER over a defined time period to mitigate specific grid or market conditions.

1.2. Program Setup and Management

Program setup and management features allow the creation, modification, completion, and deletion of programs. You create a program through the UI and via REST API, defining fields such as start and end dates, eligibility requirements, and program constraints, such as maximum events time period, maximum number of opt-outs and total dispatched energy time period.

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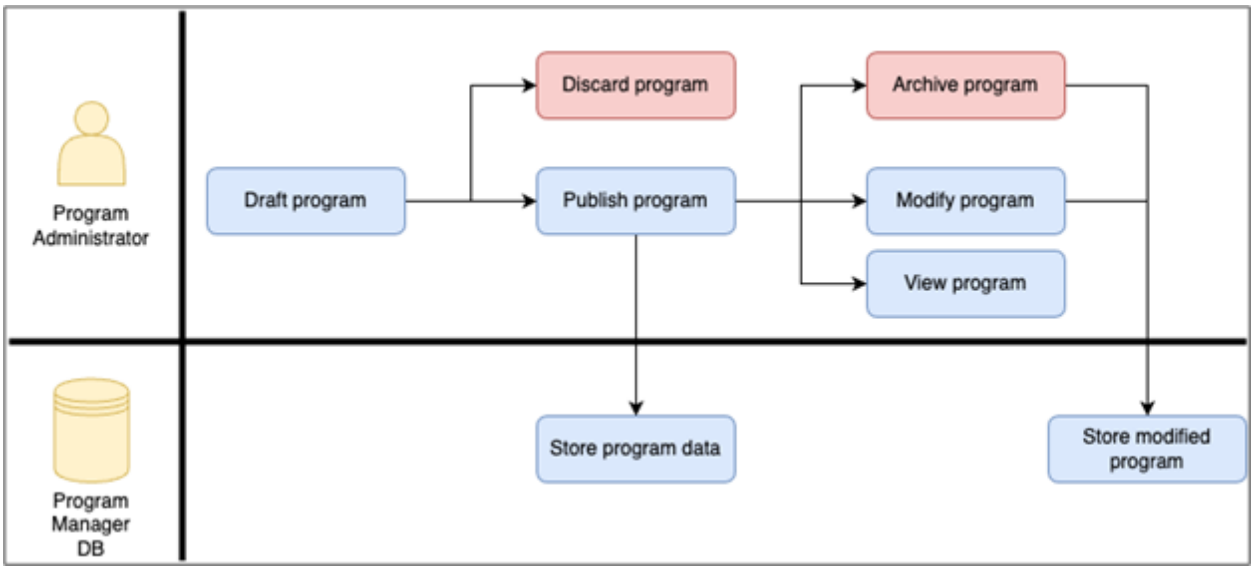

Figure 1. Program setup and management workflow

1.3. Service Provider Management

The Service Provider function allow you to create, modify, and delete service providers. Service providers can be created through the UI or by uploading multiple service providers in a .csv file via the UI or a REST API. Users can associate one or multiple DERs to a service provider, streamlining the management of various service offerings. A single DER cannot be associated to more than one service provider.

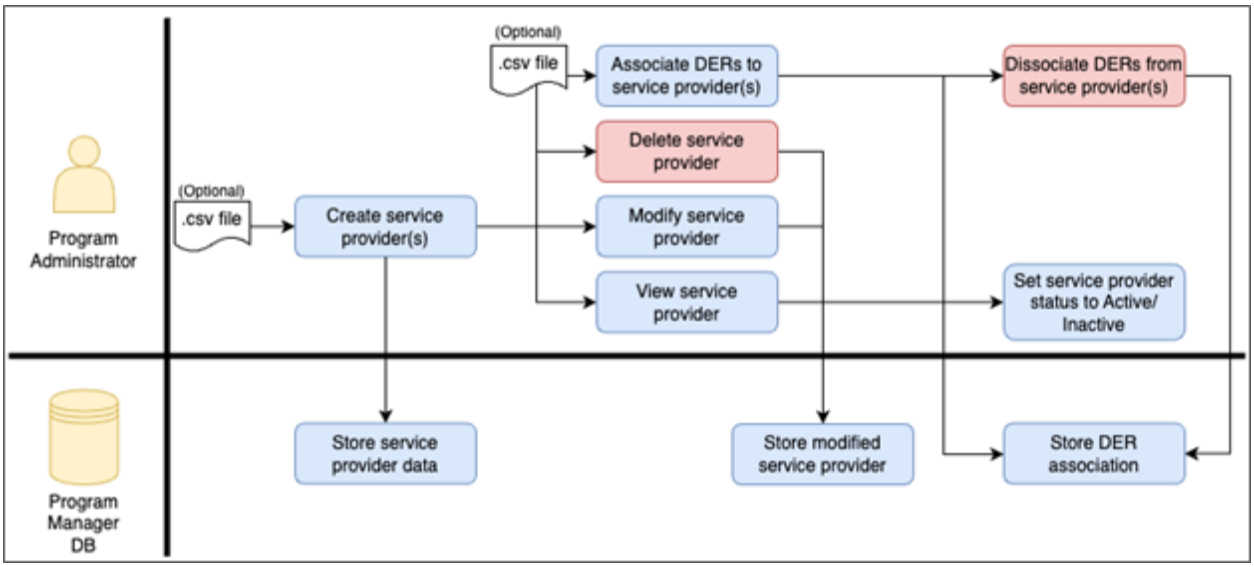

Figure 2. Service provider management workflow

1.4. DER Enrollment and Contract Management

Resource eligibility and enrollment supports multiple enrollment workflows such as internal eligibility, where users define eligibility criteria at program creation in Program Manager and external eligibility where DERs are assumed to be eligible when they are enrolled, that is, the eligibility check is done outside GridOS DERMS. This feature also allows enrollment of resources into a program through the UI or through .csv upload, and the platform records resource-to-program enrollment history. Contracts are created automatically once DERs are approved for enrollment.

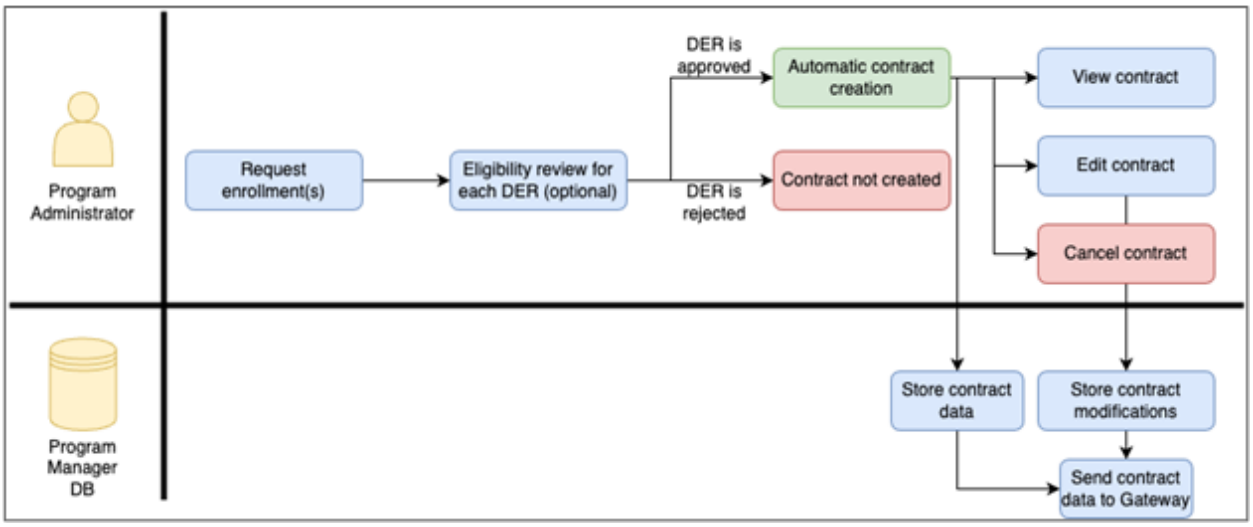

Figure 3. DER enrollment and contract creation workflow

1.5. Event Tracking and Reporting

Event Tracking and Reporting provides administrators with visibility into high-level and detailed program metrics. The system captures data from events dispatched on Gateway and displays it in Program Manager. Users can generate reports for specific dates, programs, and service providers, obtaining valuable insights such as total dispatched energy, the number of events, available capacity, and progress towards defined program constraints.

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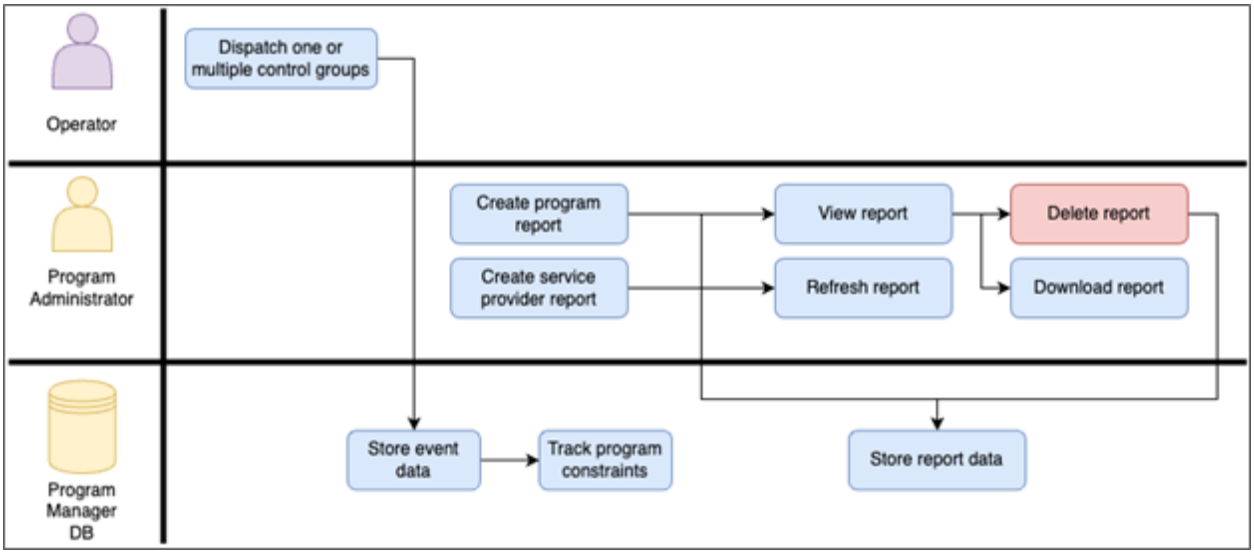

Figure 4. Event tracking and reporting workflow

1.6. High-level User and Data Workflow

A utility program administrator adds or uploads DER asset data to the DER warehouse (Model Manager) in Program Manager so that they may be associated with a service provider.

They can then set up service providers and associate DER assets to a specific service provider. A service provider with associated DER can then be enrolled in a program.

The program administrator also sets up a program, which can have a specific management objective for the grid. In creating a program, the program administrator defines the eligibility criteria of the DER to participate, defines the dispatch constraints, and defines the dispatch capacity for a given time period. Program set-up is concluded when the program is published. Once published, DER can be enrolled in the program. If they meet eligibility criteria, a contract is automatically created.

Contract data, along with program and constraint data, is sent to the Gateway for control group creation and eventual dispatch. Events are tracked by Program Manager for event tracking and program reporting.

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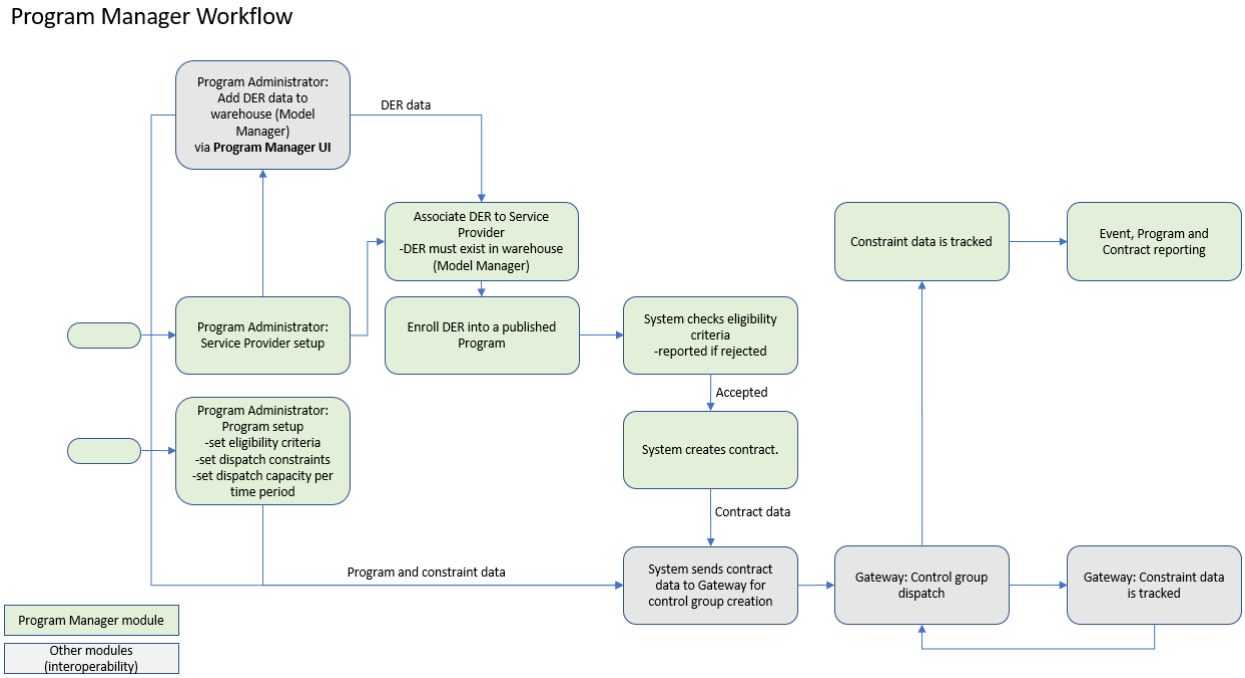

Figure 5. High-level user and data flow in Program Manager

1.7. User Persona

1.7.1. Program Administrators

Program Administrators are responsible for program implementation or management of DER programs such as EV charging pilots, smart thermostat programs, demand response programs, flex markets and direct control programs.

Key challenges for this role can be:

- Demonstrate cost-effectiveness of a pilot program
- Use multiple software products for different programs
- Silo with distribution operations
- Low quality DER data for enrollment into programs causing manual work

Program administrators look to Program Manager for its ability to integrate with current platforms and aggregators through standard protocols such as IEEE 2030.5 and OpenADR.

Assessment criteria for Program Manager would include:

1. Customer information system and customer data available
2. Logical or full-impedance network model
3. Programs need the ability to communicate through standard communication protocols

1.7.2. DSO Operations Managers

DSO Operations Managers are responsible for deciding DSO operations at a macro level, which includes choosing when to use what DER services, making decisions between options, ensuring long-term DER contracts are in place, and ensuring processes to support short-term DER contract selection are in place.

Key challenges for this role can be:

- Economic cost benefit analysis is hard to run across the network
- Information stored in different departments
- New role, constantly changing regulation on what is or is not allowed
- Tends to need to engage with DERs in markets or flexible connections contracts
- Does not have visibility of significant numbers of third-party owned DER connecting to the network, nor residential DER

Assessment criteria for Program Manager could include the Program Manager's ability to integrate with a market platform for market neutrality.

1.8. Use Cases

1.8.1. Peak Load Management

Modules required: Gateway, Model Manager, Program Manager

Demand management programs enable the flexible control of DERs to relieve grid strain during peak times, either by reducing or increasing their output.

1.8.1.1. Capabilities

Program Manager serves as the central platform for managing programs and enrollments. It allows program administrators to create and update multiple demand management programs with configurable dispatch constraints, track performance metrics, enroll DERs, and set eligibility rules to ensure only eligible DERs are enrolled. Demand Management programs allow for the aggregated control of thermostats, water heater controls, batteries, solar panels/inverters, and electric vehicle charging. Load profiles and available flexible capacity are modeled within the platform.

1.8.2. Dynamic Operating Envelopes

Modules required: Forecasting, Optimization, Gateway, Program Manager

A modern alternative to traditional interconnection processes is to allow new DER to connect to the

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grid as long as they participate in a *dynamic connection* program where they agree to respect a changing maximum export, or import, limit. These programs are called a *Dynamic Operating Envelope (DOE)* or *flexible interconnection* or *active management* programs.

In general, Dynamic Connection programs help end customers by improving PV hosting capacity, maximizing utilization and value of customer assets and ensuring equitable access to the network. It supports both end-customers and the network by reducing the burden of interconnection assessment and ensuring EV and Battery Energy Storage System (BESS) readiness. It supports the network specifically by maximizing utilization and value of network assets such as transformers and conductor, and improves the overall grid security and reliability.

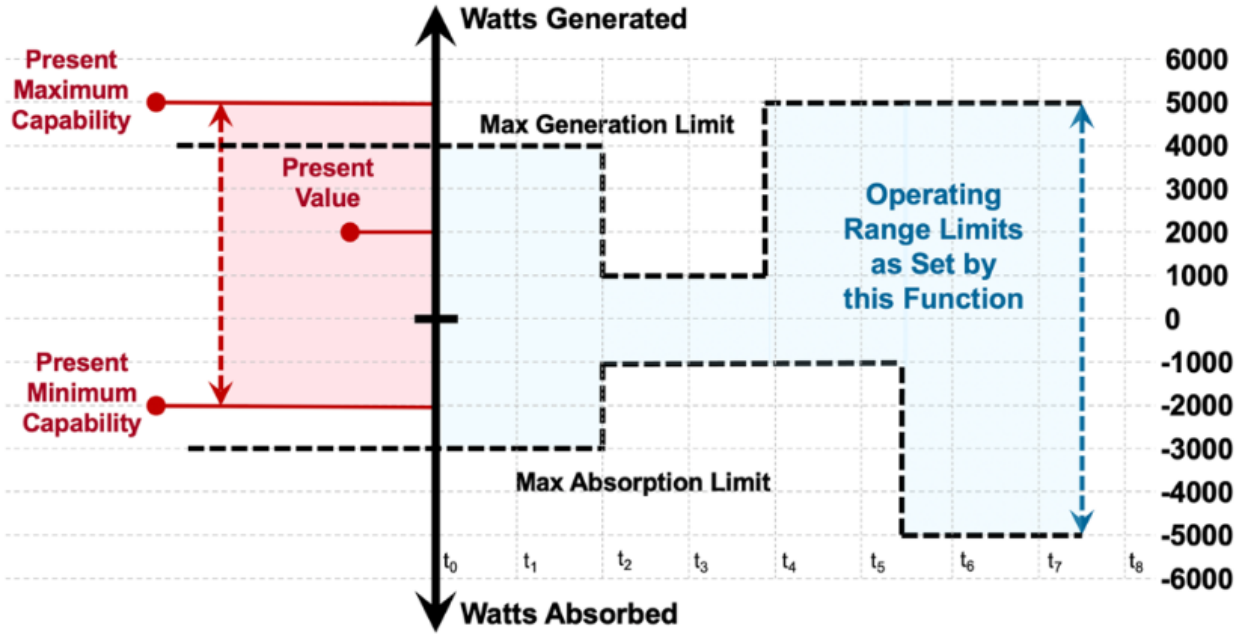

Figure 6. Dynamic Operating Envelopes (source: EPRI)

For customers that have the Forecasting, Optimization and Gateway applications deployed as part of an integrated GE Vernova GridOS DERMS solution, the DOEs can be automatically calculated based on network conditions and forecast data and communicated to DER resources using the Gateway and IEEE 2030.5.

1.8.2.1. Capabilities

DOE programs can be created in Program Manager, where program administrators can set specific parameters related to DOE. This includes limit type, calculation frequency, schedule timestep, schedule time horizon, control options, and control type. At the time of program enrollment, default limits for import and export capacity must be specified for each enrolled DER in case communication with devices fails and the default envelopes must be used. This information is sent to the Gateway through REST API for the ability to dispatch DOE-enrolled DERs.

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1.8.3. Direct Control Programs

Modules required: Gateway, Operations, Model Manger, Program Manager

Direct control programs enable utilities to control DERs through standard communication protocols such as IEEE 2030.5 or OpenADR. These programs cater to larger front-of-meter (FTM) or C&I customers, who do not require an aggregator for communications. A Power Purchase Agreement (PPA) is an example of this use case, where utilities need to monitor performance and can control the behavior of the assets in this program.

1.8.3.1. Capabilities

Direct Control programs can be created and modified in Program Manager through the “generic” program type. This program allows a program administrator to select the IEEE 2030.5 control options, as well as the dispatch options (Gateway or through API) and whether the user would like to track event data from the Gateway.

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2. Model Manager Integration

2.1. Interoperability

The *Program Manager* module uses APIs to retrieve data from external systems. You can get the schemas for the API operations used in Program Manager in these locations:

`https://<aws-url>/derms/programmanager/api/v1/system/openapi.json`

`https://<aws-url>/derms/modelmanager/api/v1/system/openapi.json`

1. Log in to Program Manager.
2. Append `programmanager/api/openapi.json` or `auditlog/api/openapi.json` or `derwh/api/openapi.json` to the URL following `/derms/` and press ENTER.
3. Copy and paste the JSON code into an API tool, and configure the payload as desired.

> **Note**
>
> `<aws-url>` refers to your environment's domain.

2.2. Data Integration with DERMS Model Manager

In order for DERs to exist in Program Manager, they must be uploaded through the DERMS Model Manager's DER Warehouse first. This requires a DER Registration file upload in .xml format through RESTful API.

The DER registration file distinguishes a DER System defined as <DER> from a DER Unit defined as <DERU>. Within the scope of this feature and in DER warehouse, it is assumed that:

- A 'DER' corresponds to a DER system and maps therefore to <DER> in the Registration file
- Each DER is associated to exactly one DER Unit (1:1 association)
- DER Units are not modelled in the DER Warehouse. Only the *type* information is gathered to be associated to the DER.

There following are the attributes used in Program Manager related to DERs. See the table below for details:

Model Manager Attribute	Program Manager Attribute	Description	Type
External ID (external_id)	DER ID	DER ID that may reference an external system.	string
Name (name)	DER Name	Free text to describe the DER for any user.	string

Model Manager Attribute	Program Manager Attribute	Description	Type
Type (der_type)	DER Type	Used to categorize the DERs by technology and function in the network. Some examples include: PV, wind, demand response, BESS and EV charging stations.	string
Rated Active Power (rated_p_nom)	Capacity	Describes the manufacturer nameplate rating of the DER.	double
Is Part of VPP? (der_vpp_type)	VPP type	Describes the type of DER, if it is grouped or not. One of: REGULAR, VPP_CHILD, VPP_PARENT	enum
VPP Parent (vpp_parent)	VPP Parent ID	If the DER is a VPP_CHILD, this field specifies the parent's DER ID	string

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3. Program Management

In program management, you create new programs, modify existing programs, or archive programs you no longer want to make available to service providers.

The **Programs** tab displays a list of all created programs, their type, the number of DER enrolled in the program, the program start and end dates, and program status.

Program Statuses

Table 1. Program Status

Status	Description
Active	The program is in effect. The start date has passed, but the program has not reached its end date.
Archived	The program is no longer available for DER enrollment. Once archived, you cannot re-activate it.
Draft	The program is in an unfinished state of set-up and has not been published yet.
Published	The program is published and available for DER enrollment.

Selecting any program in the list will open a side panel with program details, as it was configured. Program details include a program overview summary, availability constraints, dispatch information and the DER enrolled in the program. With the program details side panel open, you can enroll DER or archive the program altogether.

3.1. Program Setup

In program setup, you define the eligibility criteria for DER resources to participate in the program, define any availability requirements and define event dispatch constraints.

Program Types

A program type describes the main objective the program will achieve and allows you to set constraints that serve that objective. In Program Manager, there are 3 program types: *Demand Management*, *Dynamic Operating Envelope*, and *Generic*.

Table 2. Program Types

Program Type	Description
Demand Management	Define the demand for each DER output by setting the maximum total energy per time period.
Dynamic Operating Envelope (DOE)	Define parameters for running the DOE calculation: Limit type:Real-power (kW) or Reactive-Power (kVAR) Calc frequency: How often is the DOE calculation running. Time interval: The interval to run the DOE calculation.
Generic	A program type with no specific objective. You can select which control options you want available in the Gateway. You can also select whether this program should be dispatchable through the Gateway, through API from another system, or not dispatched at all.

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3.1.1. Creating a New Program

To create a new program:

1. Click **Add a new program** from the **Programs** tab. Follow the prompts in the program setup wizard.
2. Enter a **Program name**. Give the program a specific name that will make it easy to identify. For example, "Fleet EV Charging in District 3."

> **Note**
>
> Program names must be unique.

1. Select a **Program type** (see Program Types).
2. Provide the **Program description** which gives the detailed information for all types of programs.
3. Click **Get started**. Complete the screens presented in the setup wizard.

3.1.1.1. Step 1 General Information

1. Set **Start** and **End** dates for the program. This is the period the program will be active and DER will be dispatched.
2. Select a **Program priority**. Program priority directs the priority of conflicting controls sent to the same DER from two different programs. Priority values are: P0, P1, P2, P3, with P0 assigning the highest priority to the program you are setting up.

   > **Note**
   >
   > If two or more programs have the same priority setting, it will be up to the service provider to decide which program takes priority.

3. Select or clear **Check DER eligibility for program enrollment?** Selecting this check box will subject DER to program eligibility criteria you define. If this box is checked, you can define the minimum and maximum real power rating needed for DER to be eligible. If the checkbox is left cleared, no criteria are defined and DER will be automatically enrolled in the program when an enrollment request is created.
4. Save a draft of the setup or click **Next** to proceed.

> **Note**
>
> You can move back or ahead in the setup wizard to modify your parameters at any time by clicking the topics in the wizard sequence. Clicking **Next** or **Update** saves your changes.

3.1.1.2. Step 2 Availability

In this step, you define availability constraint parameters for DER on a per-contract basis. These are constraints that define when DER should be available, and for how long, within the contract.

Select values to define these availability constraints:

1. Select the **Operating months**: For the duration of the program, select the months you want DERs to be dispatched. This is an option if your program will be active for several years, for example, but you only want DER to be dispatched during certain months of the year.
2. Select the **Holiday exclusion**: This is an exclusion from availability. Select the statutory holidays during which you do not want DER to be dispatched. If none is selected, DER will be available on statutory holidays.
3. Select **When should DER be available?**: Select either **All the time (24/7)** or **At particular service windows**. If for a service window, define the service window by selecting a day of the week, then select start and end times for those days. You can add multiple exclusive service windows, if needed.
4. Save a draft of the setup to date or click **Next** to proceed.

3.1.1.3. Step 3 Dispatch

In this step, you define event constraints on the dispatch of a DER on a per-contract basis. For example, you may want to limit the number of times a DER can opt out of participating in the contract. You can also set limits on the amount of time a DER will be available during an event (single event duration). You can set limits on the amount time a DER will be available across all events (cumulative event duration). Finally, you can define the maximum number of events a DER is subject to.

> **Note**
>
> An instance of opting out means the DER or service provider is allowed, without penalty, to abstain from responding to a dispatch.

When a dispatch is created and later cancelled with a status of "CANCELLED," the system handles it differently based on when it is cancelled:

- If the dispatch is cancelled before its scheduled start time:
  - It is treated as if it never occurred.
  - It is not included in any capacity or budget calculations.
  - It does not appear in event- or dispatch-related reports.
  - It is not counted in any widgets or summaries that show the number of events.
- If the dispatch is cancelled after its scheduled start time:
  - It is treated as if it ran for its full scheduled duration.

- Any remaining or leftover capacity or budget is adjusted accordingly. Select limits to define dispatch constraints for these parameters.

Opt-Outs

a. **Maximum opt-outs per contract**: Enter a number of permitted opt-outs by a DER over the contract's time period.
b. **Opt-outs time period**: Select the period (day, week, month, or year) during which opt-outs are permitted and will be counted.

Single Event Duration

This constraint is a minimum or maximum limit on the amount of time a DER is dispatched in a single event. For example, if you define a maximum duration of 6 hours, then the contracted DER cannot be dispatched for more than 6 hours in a single event.

Use a minimum and/or maximum constraint parameter to set upper and lower limits.

a. **Single event duration**: Select either a **Maximum** or **Minimum** parameter that will define the duration that the DER must be available during an event.
b. **Minutes**: Enter the number of minutes, as a minimum or maximum, a single DER can be dispatched during an event.
c. Click **Add another** to define an additional constraint.

Cumulative Event Duration

This constraint is a minimum or maximum limit on the amount of time a DER is dispatched across all events in the contract. For example, if you define a minimum cumulative event duration of 10 hours, then the contracted DER must be dispatched for at least 10 hours cumulatively in the time period specified.

Use a minimum and/or maximum constraint parameter to set upper and lower limits.

a. **Cumulative event duration**: Select either a **Maximum** or **Minimum** parameter that will define the duration that the DER must be available across multiple events in a period.
b. **Minutes**: Enter the number of minutes, as a minimum or maximum, a DER can be dispatched during the period.
c. **Per time period**: Select the period for which this constraint applies.
d. Click **Add another** to define additional constraints.

Maximum Number of Events

a. **Maximum number of events**: Select a maximum number of dispatch events in a period a DER is subject to in the contract.
b. **Per time period**: Select the period for which this constraint applies.

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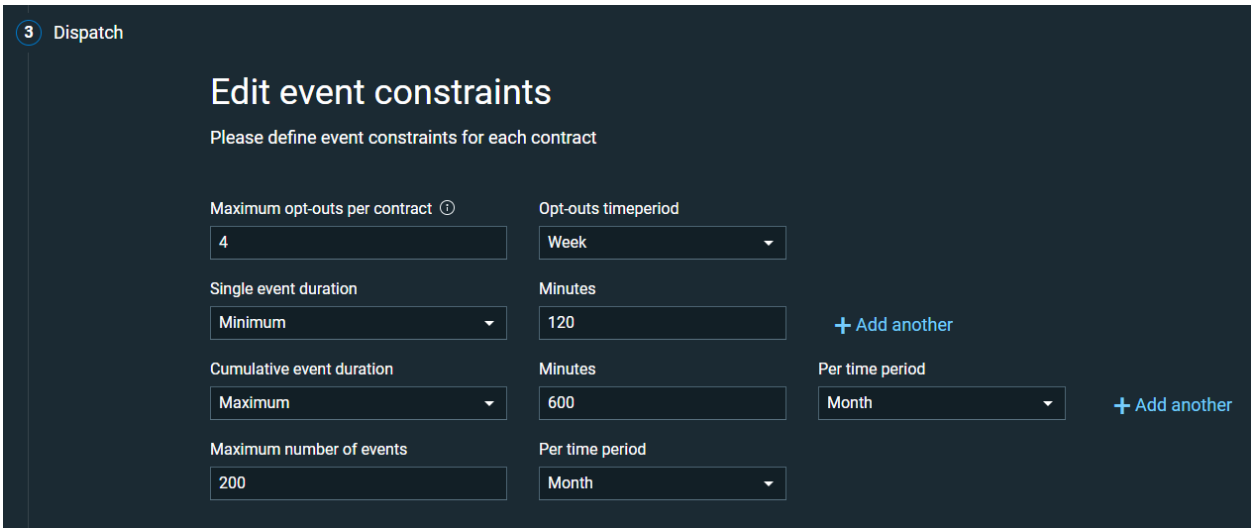


Figure 7. Dispatch event constraints

Save a draft of the setup to date or click **Next** to proceed to the next step in program creation.

3.1.1.4. Step 4 Program

In this section of the setup, you define options and limits specific to the program type you are creating by dispatching DERs to increase or decrease output or consumption.

Demand Management Programs

Demand management programs allow you to control energy output or consumption on the grid at certain times or locations in response to grid conditions to increase or decrease demand. You can define the limits for DERs by selecting values for the following parameters.

If your program type is for **Demand Management**, select these program options:

1. **Maximum total energy per time period**: Enter the maximum value of energy allowed for the time period.
2. **Unit**: Select a unit value for your constraint (kWh or MWh)
3. **Time period**: Select the time period this constraint valid for (day, week, month, or year).

Dynamic Operating Envelope Programs

Dynamic operating envelope programs allow you to control power limits that each DER can import or export to the grid in a given time interval to manage DER variable load and increase Quality of Service (QoS). You can define the limits for DERs by selecting values for the following parameters.

If your program type is **Dynamic Operating Envelopes**, select these program options:

1. **DOE limit type**: Select Real-power or Reactive-power
2. **DOE calculation frequency**: Select frequency value from the drop-down menu for how often the DOE calculation should run.

3. **DOE schedule timestep (minutes)**: Enter a time interval, in minutes, for the DOE schedule. This specifies the length of interval the DOE calculation should be provided.
4. **Control options**: CSIP or CSIP-AUS

> **Note**
>
> CSIP (Common Smart Inverter Profile) is a standardized communication protocol for smart DERs, such as solar photovoltaic (PV) systems, battery storage, and electric vehicles. CSIP-AUS is an adaptation of CSIP for the Australian context and is tailored to the specific characteristics of the Australian National Electricity Market (NEM).

5. **DOE Schedule time horizon**: Enter or select a total time duration the DOE will be scheduled for.
6. **Time Period**: Select the unit of time for the time horizon (minutes, hours, or days).

In this example, the DOE program constraints would be described as follows: For a 24-hour time horizon, the DOE program is calculating the active power limits at 5-minute intervals. The communication of dynamic limits to each DER is through CSIP instructions.

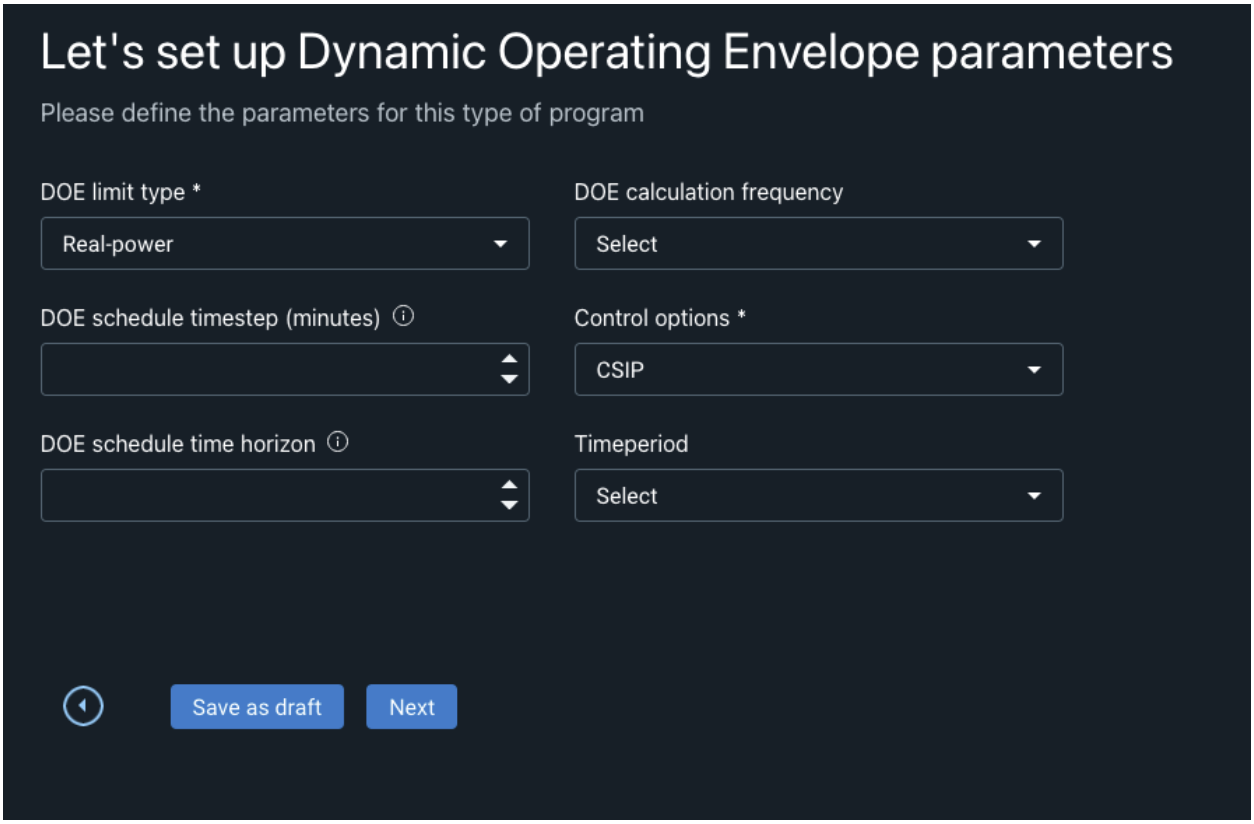


Figure 8. DOE program settings example

Generic Programs

A Generic program allows you to build a program with control options.

> **Note**
>
> The controls offered are based on the IEEE 2030.5 standard.

1. **Control options**: Select one or more control options from the drop-down menu.
2. **How will this program be dispatched?**: Select the method of dispatch (**Gateway**, **API**, **Other**)

   > **Note**
   >
   > If the program is not to be dispatched, select **Other**. If the program is to be dispatched through a system other than Gateway, select **API**.

3. **Do you want to track event data from the Gateway?**: Select **Yes** or **No**. If **Yes** is selected, event data is reported and displayed in Program Manager.

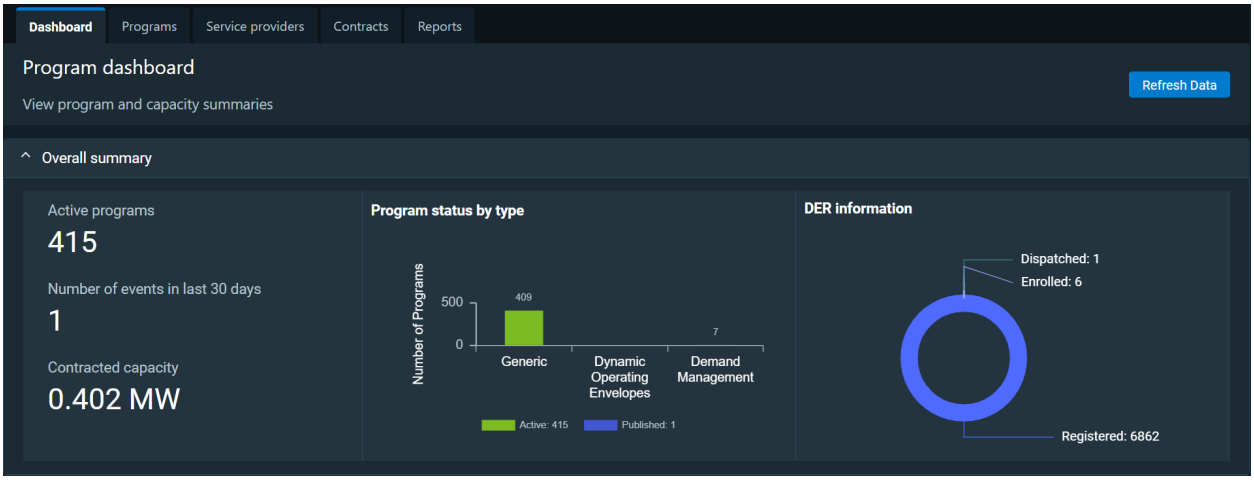
3.1.1.5. Step 5 Publish

In this section of the setup, review your program configuration. You can edit any of the selections made by clicking the edit icon , save the program as a draft to resume later or publish the program.

When ready, click **Publish program** to publish the new program. Publishing sets the new program status to **Published** and adds it to the list on the **Programs** tab. Publishing a program makes it available to service providers for enrollment.

4. Program Dashboard

The Overall summary provides a high-level insights of program and DER activity, including active programs, events in the last 30 days, total contracted flexible capacity, and recent DER usage



- **Active programs** provides the total number of programs which are in Active status. Programs are in Active status when their start date is in the past and their end date (if defined) is in the future.
- **Number of events in the last 30 days** provides the dispatched events through DERMS Gateway from Demand Management and Generic programs. Multiple contracts can be dispatched in a single event.
- **Contracted capacity** provides the flexible down capacity for all Demand Management programs with committed down capacity. This is the total megawatts which can be curtailed in demand if the user needs to hit the “Big Red Button”.
- **Program status by type** graphically represents the number of Active and Published programs by program type.
- **DER information** graphically shows the DER usage, including the total number of registered DERs in DERMS Model Manager, the number of DERs enrolled into programs, and the number of DERs which have been dispatched, enrolled & registered in the past 30 days.

4.1. Program Capacity Visualization

- Program-level visibility into the hourly availability of DERs enrolled in Demand Management programs, enabling DERMS Operators to manually dispatch DERs when needed to resolve grid constraints.
- The system calculates and stores the aggregated available "flex down capacity" (in MW) for each active program across a 24-hour period. It considers multiple availability constraints at both the contract and program levels and is updated hourly. Data is made accessible via a backend API for integration and operational use.

- Contracted flexible capacity section let the user controls which programs are shown, for which dates, and which type of capacity is displayed. User can select dates up to 10 days in the future. The same filters apply to both Matrix View and Graph View.

4.2. Filter By Date

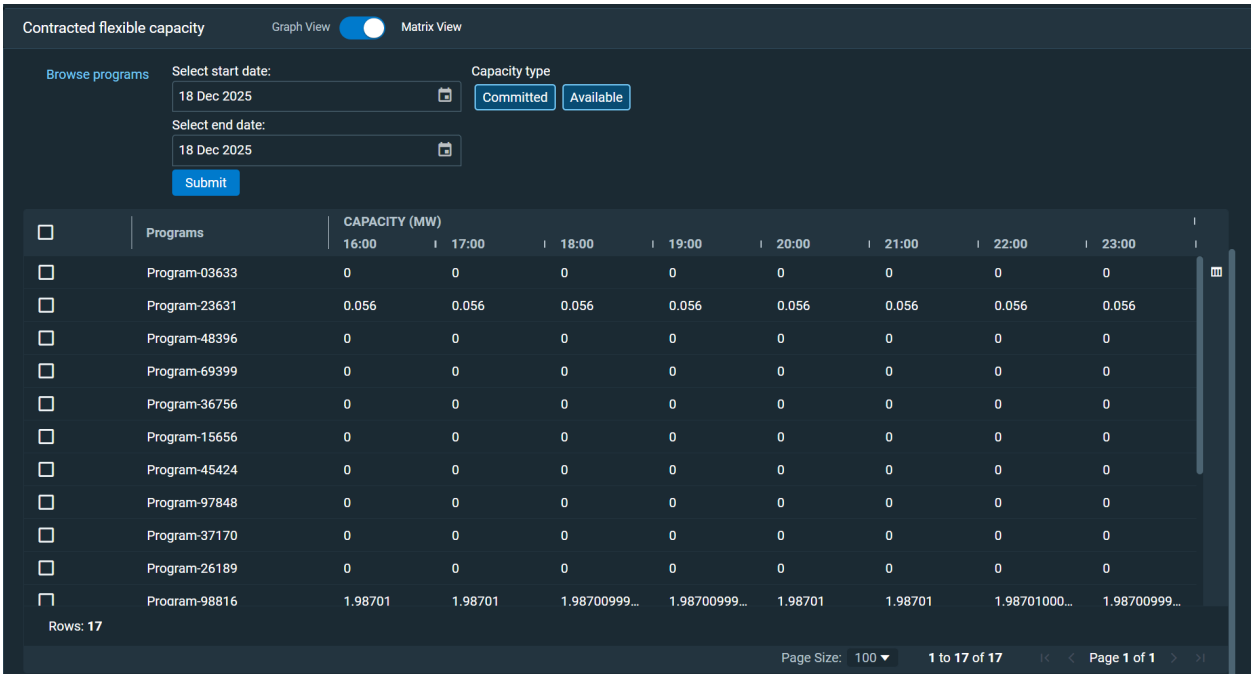


Figure 9. Default matrix view of a day

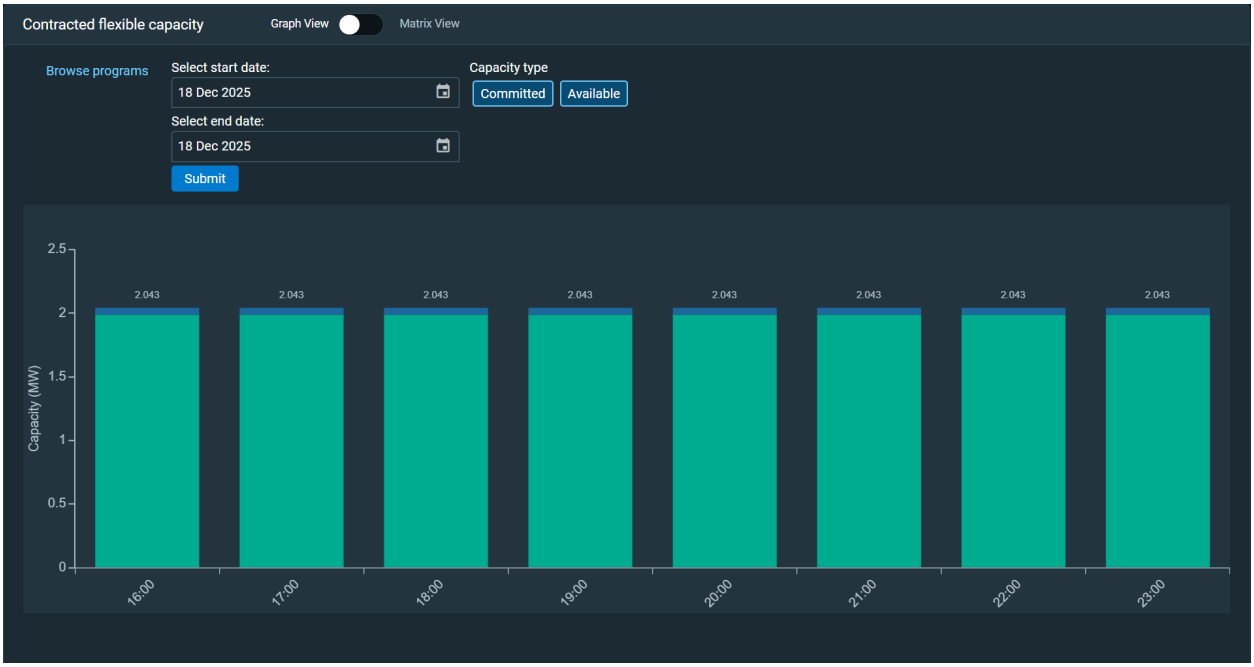


Figure 10. Default graph view of a day

- This displays hourly columns based on the "start datetime" values in the availability data.
- Corresponding hourly availability is mapped to "sum_flex_down_capacity" for each program.

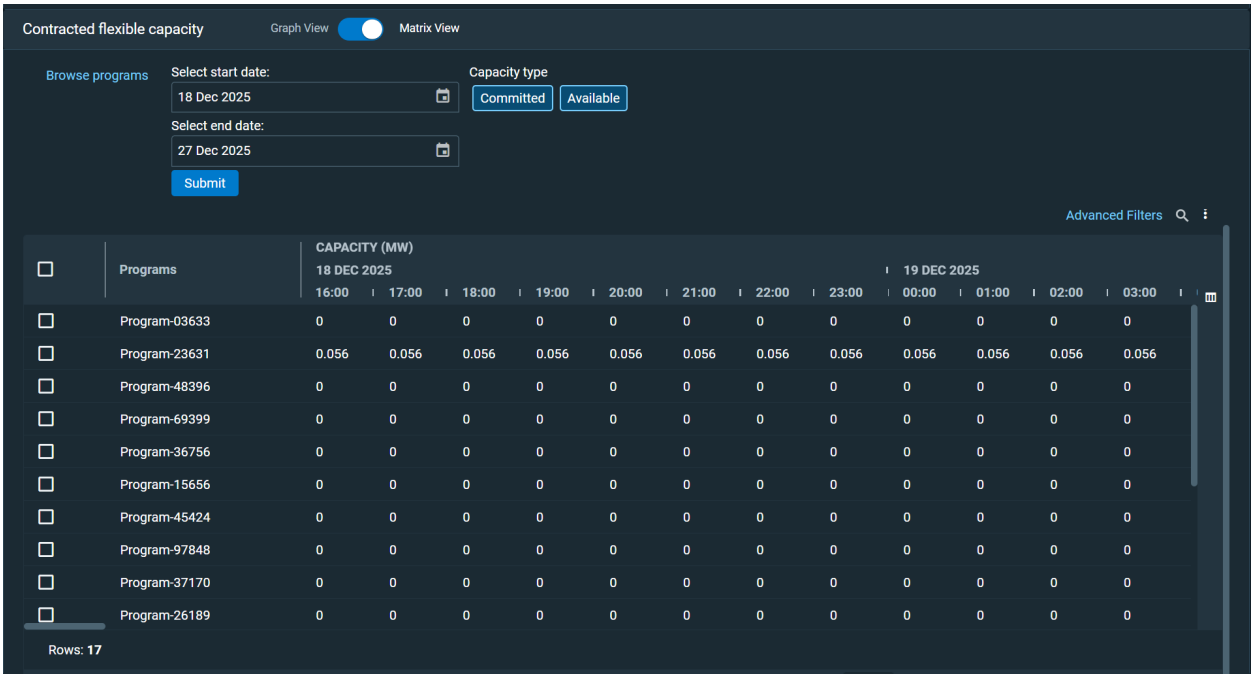


Figure 11. 10 days matrix view

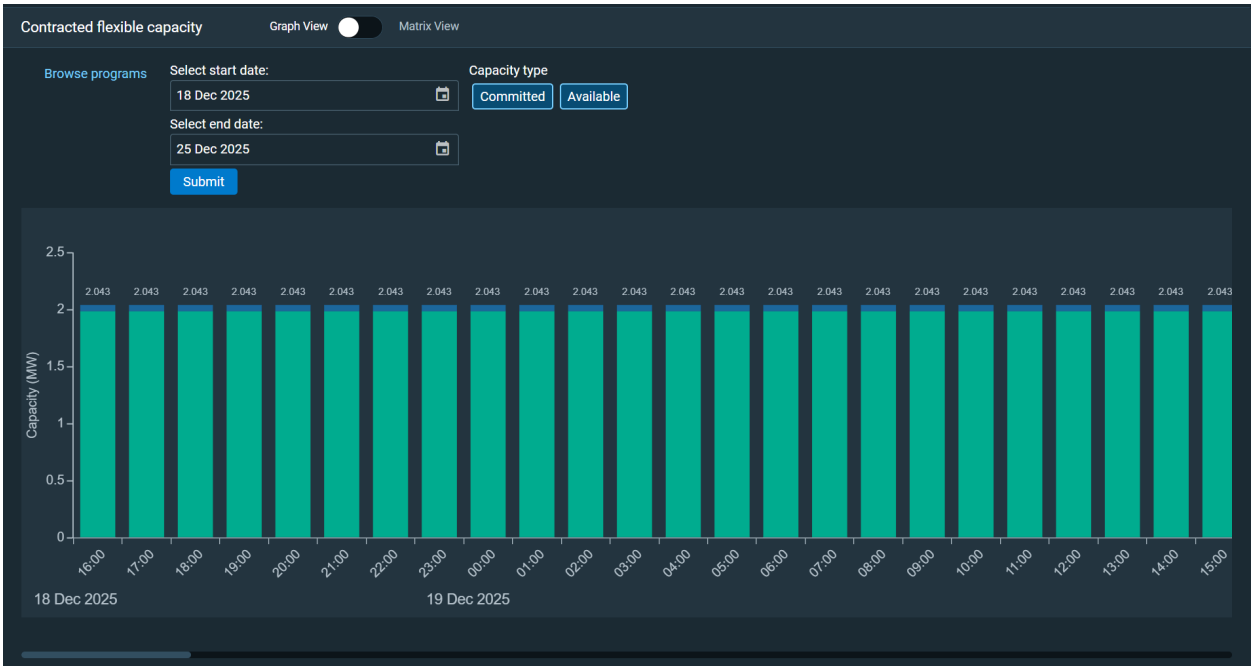


Figure 12. 10 days graph view

4.3. Filter by Program Name

- Selecting **Browse program** displays the list of Demand Management programs, which on select filters the respective programs in matrix and graph view.

Figure 13. All Programs

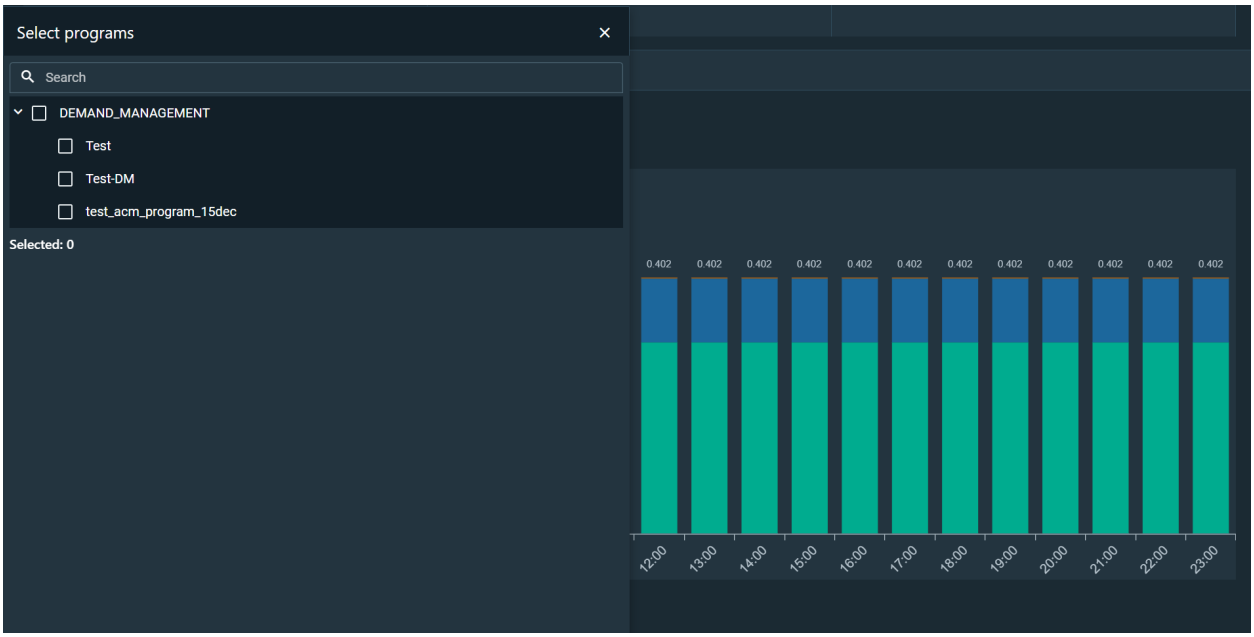


Figure 14. List of Demand Management Programs

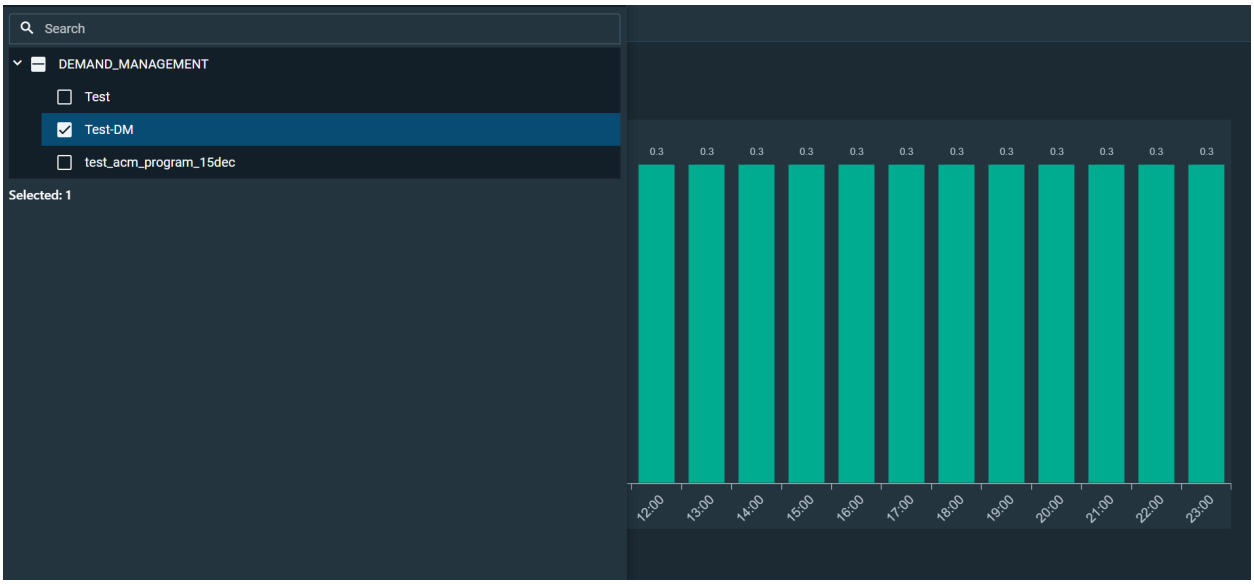


Figure 15. Filtered programs

4.3.1. Filter by Capacity type

- By default total contracted capacity (sum of available and committed capacity) is displayed which can be filtered on the basis of **Committed** and **Available** capacity type and will be updated in both matrix and graph view.

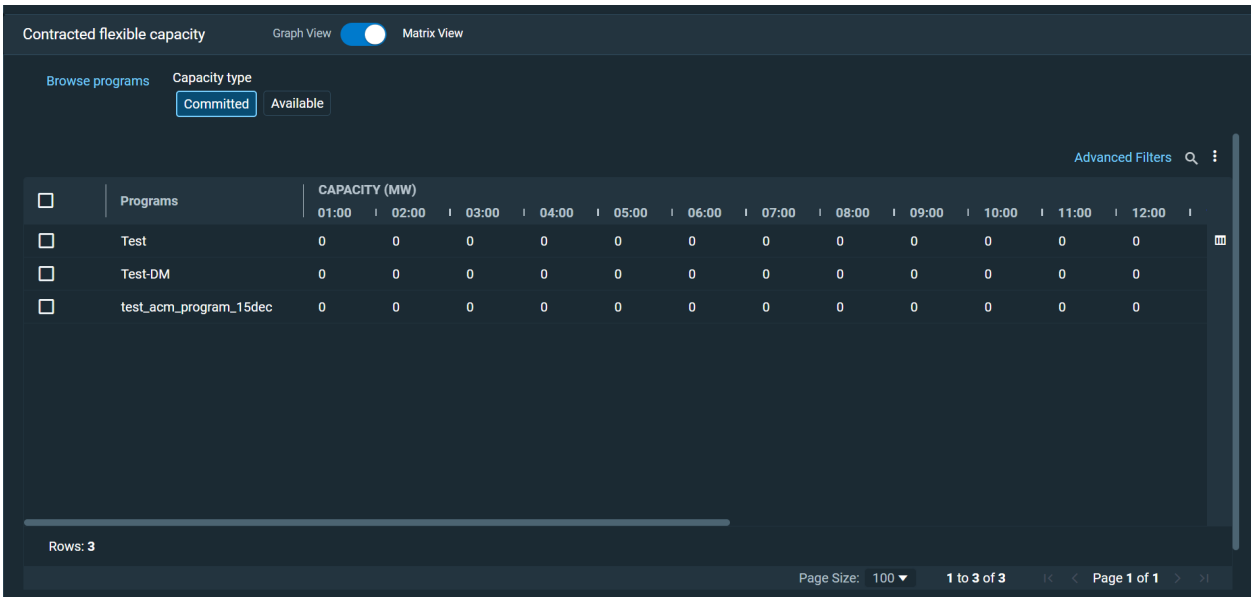


Figure 16. Committed Capacity Type

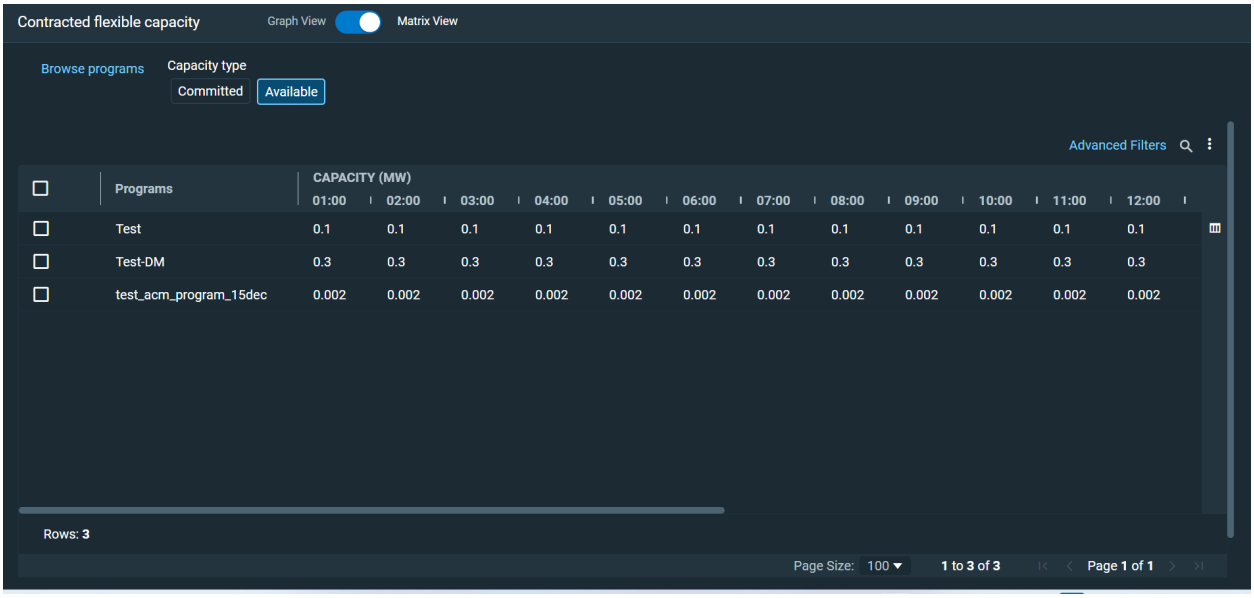


Figure 17. Available Capacity Type

5. DER Enrollment

5.1. Enrolling DERs in a Program

> **Note**
>
> Program Manager references a database of DER in Model Manager for the DER assets to be enrolled in a program. DER data is uploaded to Model Manager during the initial implementation of DERMS by your GE Vernova Professional Services team.

To enroll DER in a program:

1. Locate the program on the **Programs** tab.
2. From the side panel details, click **Enroll DER**. The **Enroll DER** tab opens.

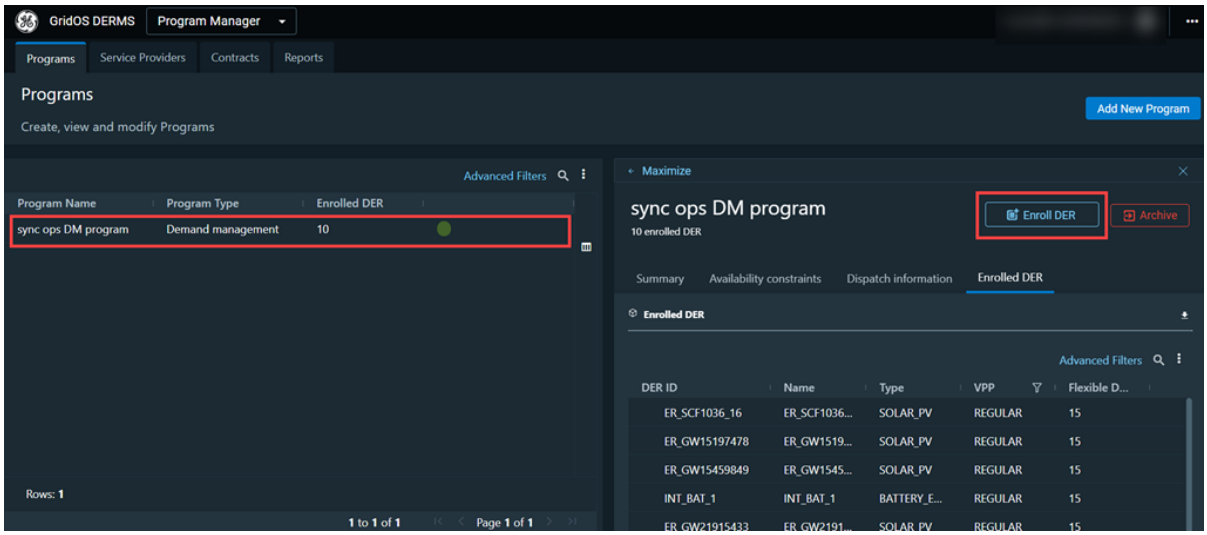


Figure 18. Enroll a new DER into a program

3. Select a **Manual Enrollment Request** to select a single DER from a dropdown list of DER assets in the system. Select **Multiple Enrollment Request** to upload a .csv file of multiple DER asset IDs at once to have multiple DERs in the system enrolled in the program.

   > **Note**
   >
   > Download and use the program specific template to ensure you are uploading a data file that meets data format requirements. Once uploaded, the DER are enrolled in the program and associated with the service provider using the program.

4. Click **Submit**.

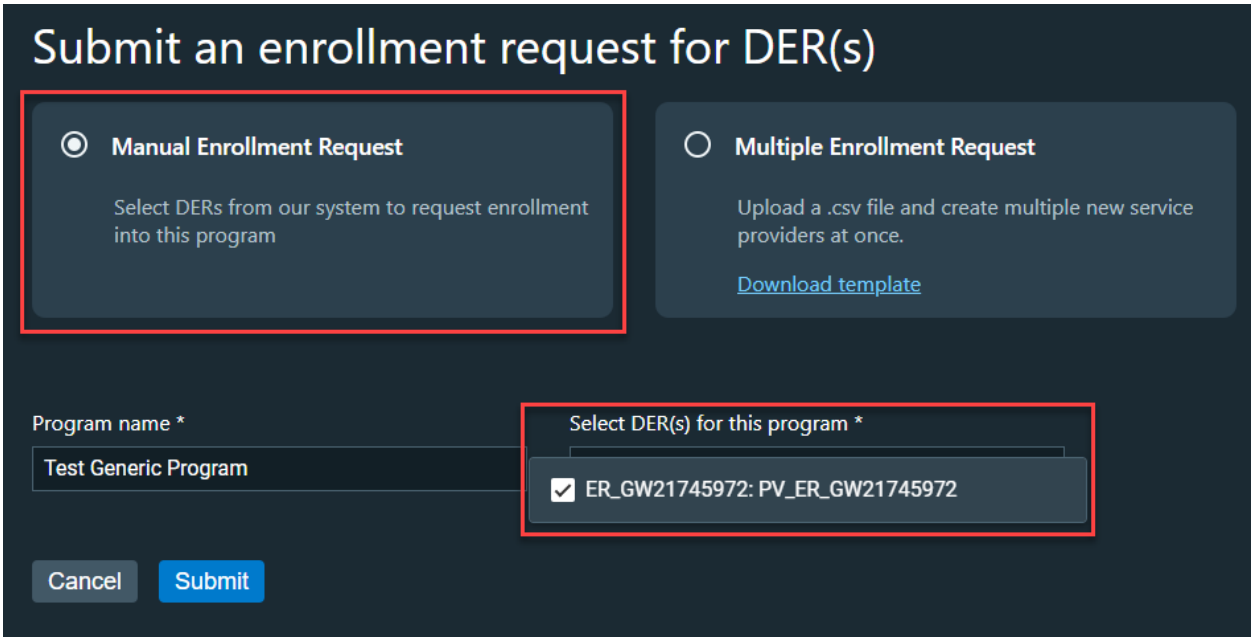


Figure 19. Manually enroll one or more DER into a program

5.1.1. Enrollment in Generic Program

- When enrolling assets in a **Generic Program**, the system manages **curve-based controls** (such as Volt-Var or Frequency-Watt) by selecting the appropriate Curve ID from the dropdown to associate the DER with the required grid-behavior.
- If a program was created without any curve-based controls, curve options will be hidden and cannot be added later during program modification.
- If a program includes at least one curve-based control, it enables to view, select, and modify both curve and non-curve control options as needed.

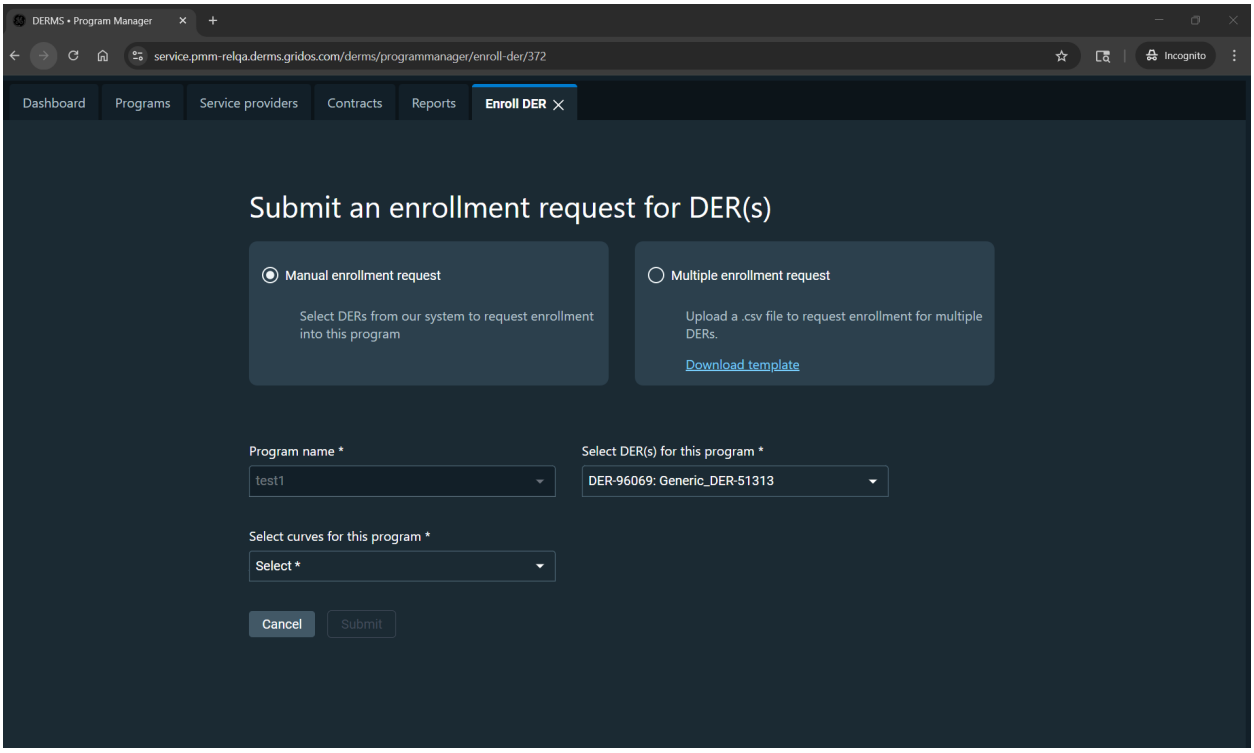


Figure 20. Enrollment in Generic program

5.2. Enhanced Navigation in the Enrolled DER Table

- To manage your assets more efficiently, the Enrolled DER table now includes direct links to related entities:

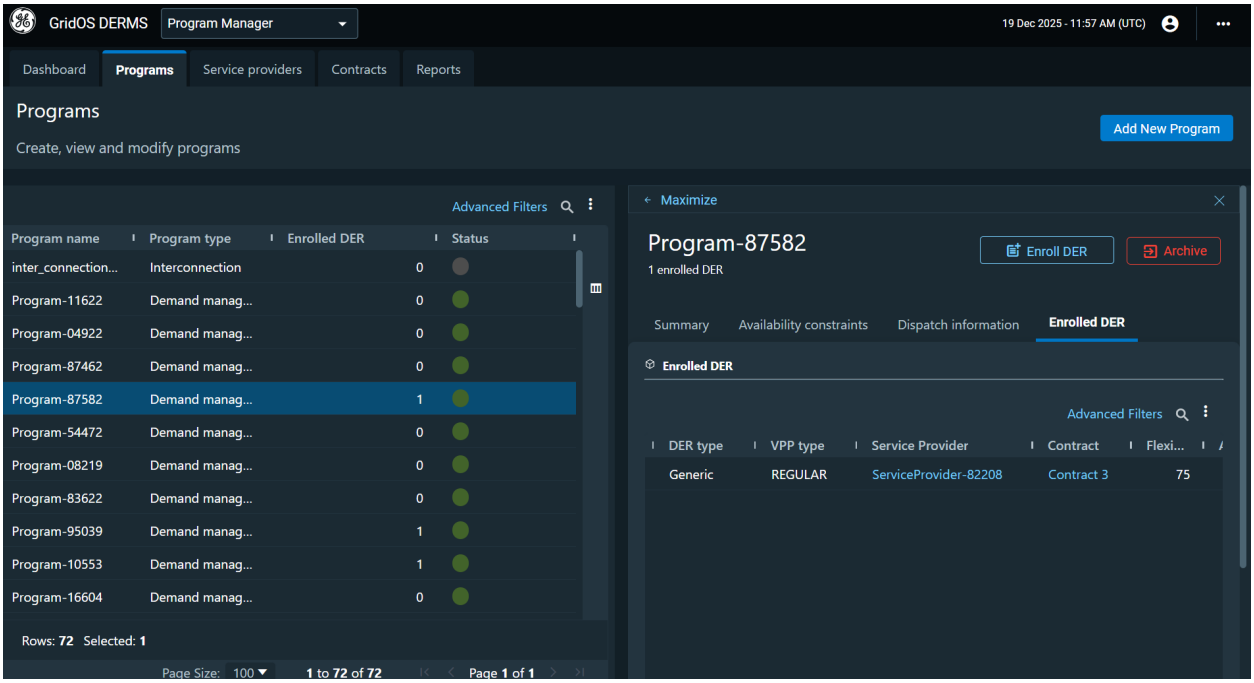
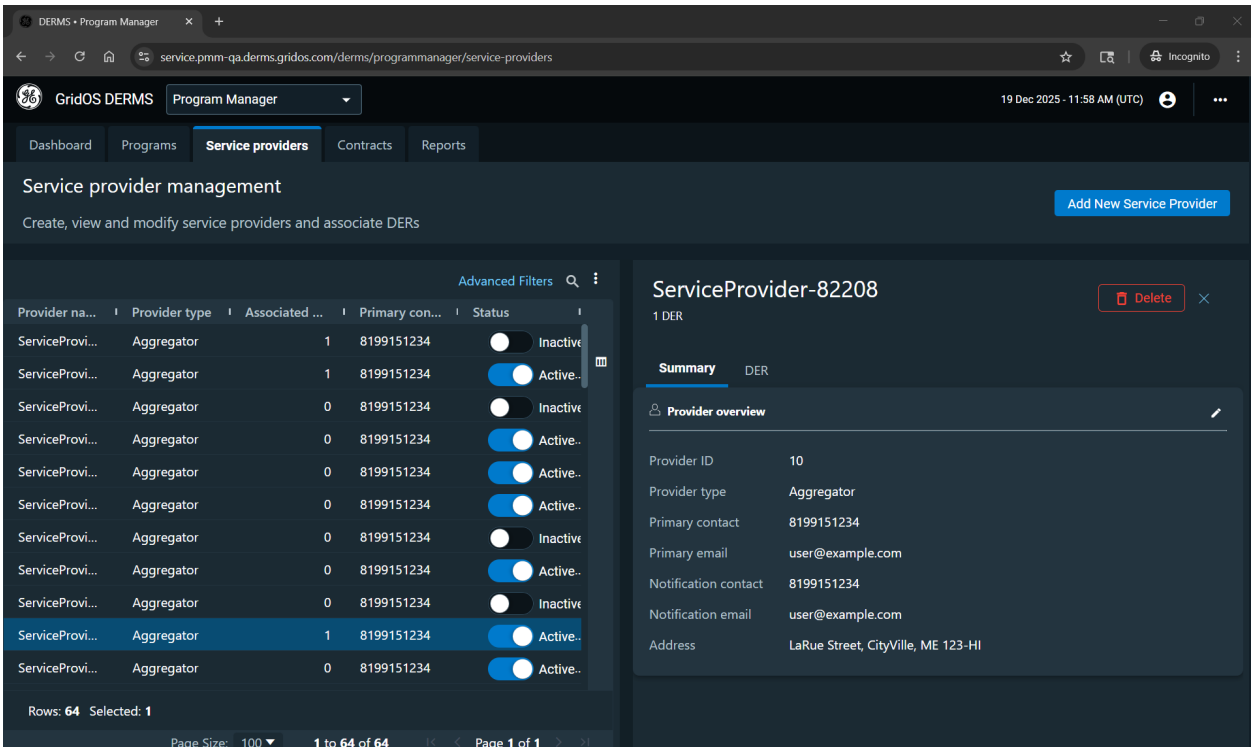


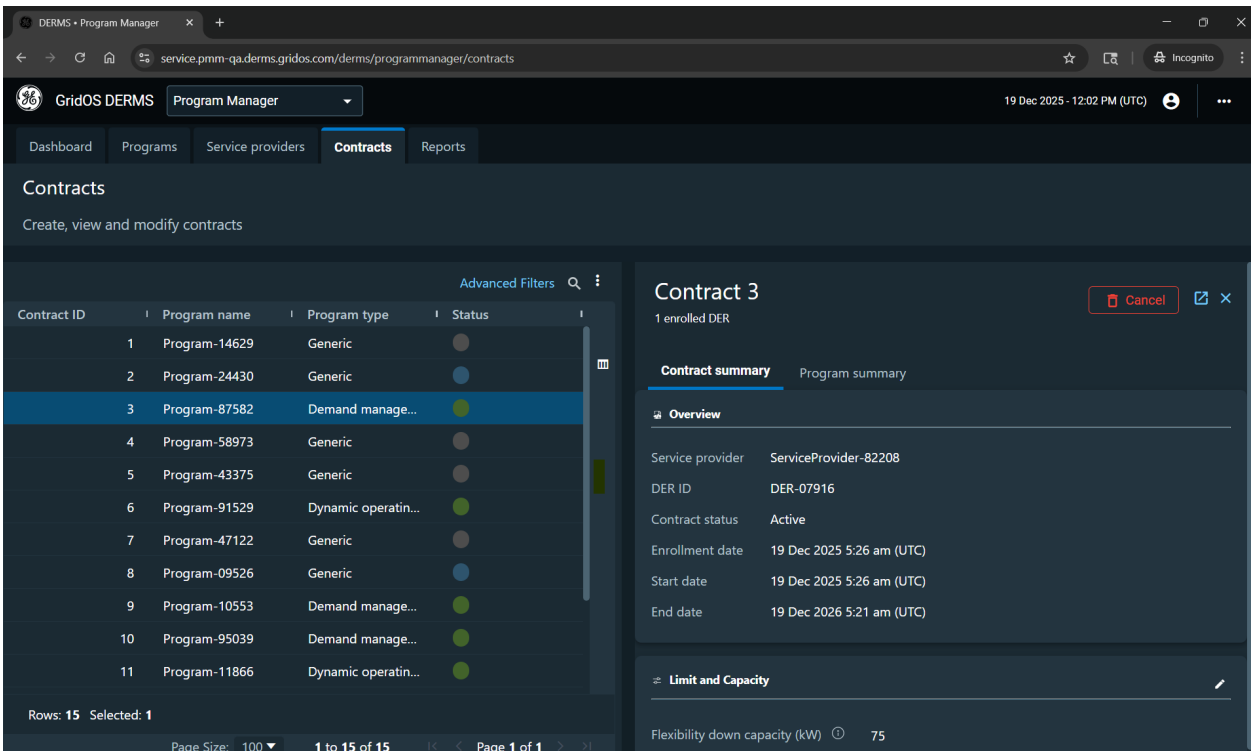
Figure 21. Enrollment in Generic program

- **Service Provider**: View the associated service provider name. Clicking the link will open the Service Provider tab in a split-screen view for quick reference.



- **Contract**: View the specific contract name associated with the enrollment. Clicking the link navigates you to the Contracts tab, opening the details in a split-screen view so you never lose

your place in the enrollment list.



5.3. Modifying a Program

You can modify any parameters of an existing program in any status, including enrolling new DER or deleting an enrolled DER. You can also archive the program, making it unavailable for enrollment, unavailable for dispatch, or for further modification. Archiving a program cannot be undone.

To view and then modify a program, select it from the list on the **Programs** tab to open the side panel. From the side panel details, click a tab to view the program's **Summary**, **Availability Constraints**, **Dispatch Information**, and **Enrolled DER**. From any of these tabs, click the edit icon in the section to open the **Edit program** tab, modify the configuration options, and click **Update** to save the changes.

6. Service Provider Management

The Service provider management area of the module allows you to view, create, modify, or delete service providers. Once a service provider is added, you can associate one or more DERs to it.

> **Note**
>
> In this section, Program Manager requires some PII (Personally Identifiable Information) to be collected for a service provider, namely a name, email address, and phone number. Please be aware of your responsibility to maintain appropriate practices for data privacy and security. See GDPR Letter of Applicability.

Select any service provider from the table to view the provider's details in the side panel. Side panel details display a **Summary** tab with a provider overview, and a DER tab displaying the associated DER.

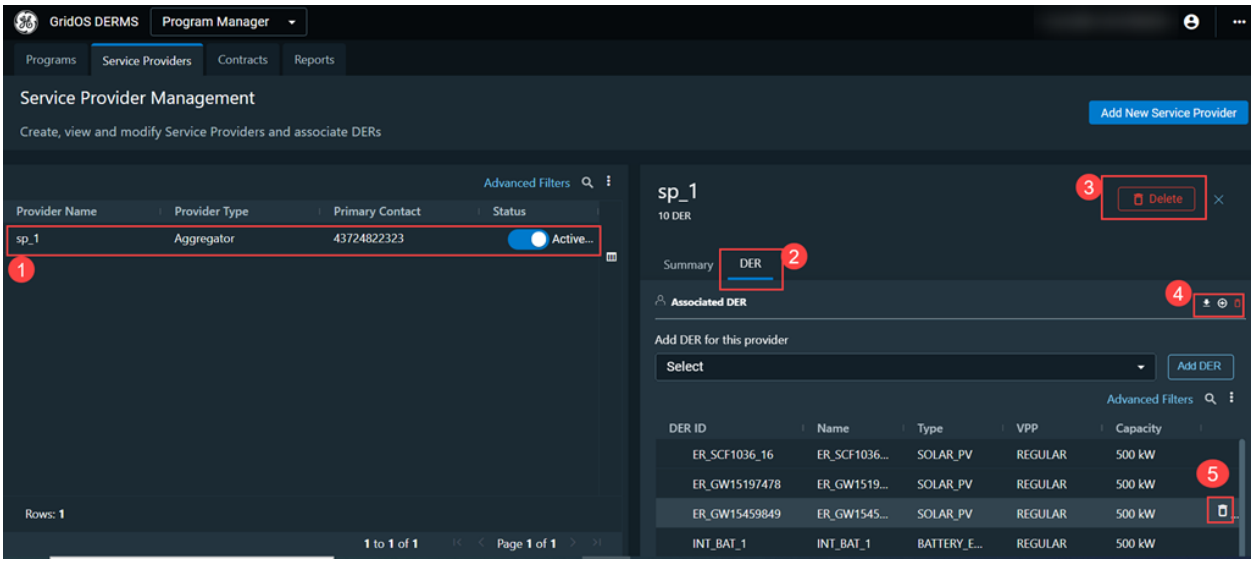


Figure 22. Service provider area functions

1. List of service providers. Selecting a service provider opens a details side pane.
2. The DER sub tab shows all DERs associated with the selected service provider.
3. The Delete icon deletes the service provider and all associated DERs from the module. Any contracts for enrolled DER are automatically cancelled.
4. The DER records can be managed by using a Download, Add, or Delete icon. Download provides an Excel list of all DER records associated with the selected service provider, and includes DER ID, DER name, DER type, DER nameplate rating, rating unit, and resource type. The Delete icon removes all DER associations with the selected service provider.
5. The delete icon that appears next to a single, selected resource deletes only that resource. This action cannot be undone.

6.1. Service Provider Setup

In Service provider setup, you can add a new service provider one at a time on the **Service providers** tab or add multiple records through a .csv file upload to Program Manager.

> **Note**
>
> If you want upload multiple service providers using csv file upload, a file template is available.

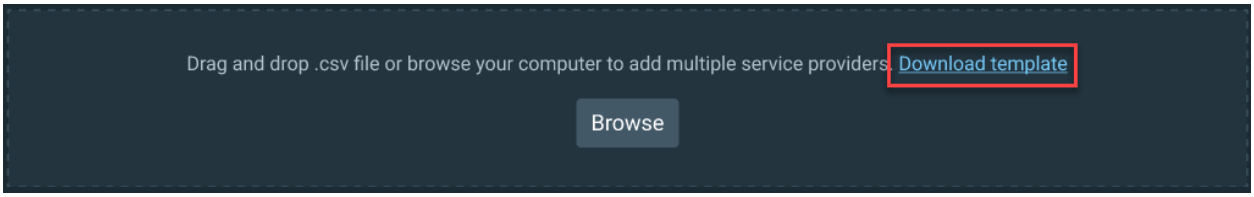


Figure 23. On the **New provider** *tab, browse to upload multiple service providers. Use the attached template to understand file requirements.*

6.1.1. Creating a New Service Provider

Click **Add a new service provider** to create a new record. A New service provider tab opens.

To create multiple new service providers at one time, see Creating New Service Providers Using File Upload

To create a new service provider, see Creating a Service Provider Manually

6.1.1.1. Creating a Service Provider Manually

To create a single new service provider manually, complete these fields:

1. **Provider name**: Enter a name for the provider.
2. **Provider type**: Select one of **Aggregator**, **C&I**, **Residential** or **Utility**.

   > **Note**
   >
   > Aggregators are DER owner or operator entities that act as intermediaries between DERs and the power system. A service provider is an aggregator if they can pool DERs together to provide flexibility.This is in contrast to a Residential service provider, who is likely the owner of DERs behind the meter at a single residential location.

3. Click **Get started**. The setup wizard starts.

6.1.1.2. Step 1 General Information

Enter general contact information for the service provider.

1. **Primary contact**: Enter a primary 10-digit phone number for the provider.
2. **Primary email**: Enter a valid email address for the provider.
3. **Notification contact**: Enter a phone number for dispatch notification purposes.
4. **Notification email**: Enter a secondary email address for dispatch notification purposes.

> **Note**
>
> Notifications are custom to a deployment and do not come off-the-shelf.

5. Complete the remaining address fields for the service provider.
6. Click **Next**.

6.1.1.3. Step 2 Resources

Add one or more DER resources to be associated with the service provider you are creating. You can skip this step in the setup wizard and associate DERs at a later time.

Associate a single DER to a service provider

To associate a single DER to a service provider:

1. Select a DER from the dropdown list.

> **Note**
>
> If you associate a single DER using the dropdown selection, you must have already added it to the list of DER for it to be available for selection. See Enrolling DERs in a Program.

2. Click **Next**.

Associate multiple DER to a service provider at one time

You can upload DER data and at the same time associate them with the service provider you are creating.

> **Note**
>
> The template file has column headers and data format requirements.

To upload a file:

1. Click **Browse** to open a .csv from your drive (or drag and drop a file to this area).
2. Click **Next**.

6.1.1.4. Step 3 Publish

In this final step, review the service provider details and associated DER to ensure they are correct.

Edit any details by clicking the edit icon from either section.

When ready, click **Publish** to publish the new service provider. Publishing sets the new service provider status to *Active* and adds it to the list on the **Service providers** tab.

6.1.1.5. Creating Multiple Service Providers Using File Upload

To create multiple service providers, you can either upload a .csv file through the UI or upload via a REST API call.

6.1.1.5.1. Uploading a .csv file

To upload a file, click **Browse** to open a .csv from your drive (or drag and drop a file to this area). Follow the steps presented by the setup wizard.

> **Note**
>
> The template file has column headers and data format requirements.

6.1.1.5.2. Uploading via API

To upload a file via API, you must use an API tool, such as Postman, to call an appropriate API. You can view endpoints for service provider APIs here: https://<aws-url>/derms/programmanager/api/openapi.json

7. Contract Management

As DERs are enrolled into a program and should pass all eligibility criteria, a contract is automatically created. Contracts cannot be manually created.

The **Contracts** tab of the module allows you to view, modify, and cancel contracts.

Select any contract from the table to view the contract summary, include what status the contract is in, and program summary details in a side panel.

Contract Statuses

Status	Description
Active	The contract is currently operational and between its start and end dates. You can modify contract details even while the contract is active.
Cancelled	The contract was cancelled and is no longer active. You can undo a cancellation if it was manually cancelled. You cannot undo a cancellation if the contract was automatically cancelled.
Expired	The contract is no longer active because it has passed its end date.
Accepted	The contract's start date is imminent.

The Contract summary tab side panel displays an abridged view of contract details (on the Contract summary tab) and an abridged view of the program details (on the Program summary tab).

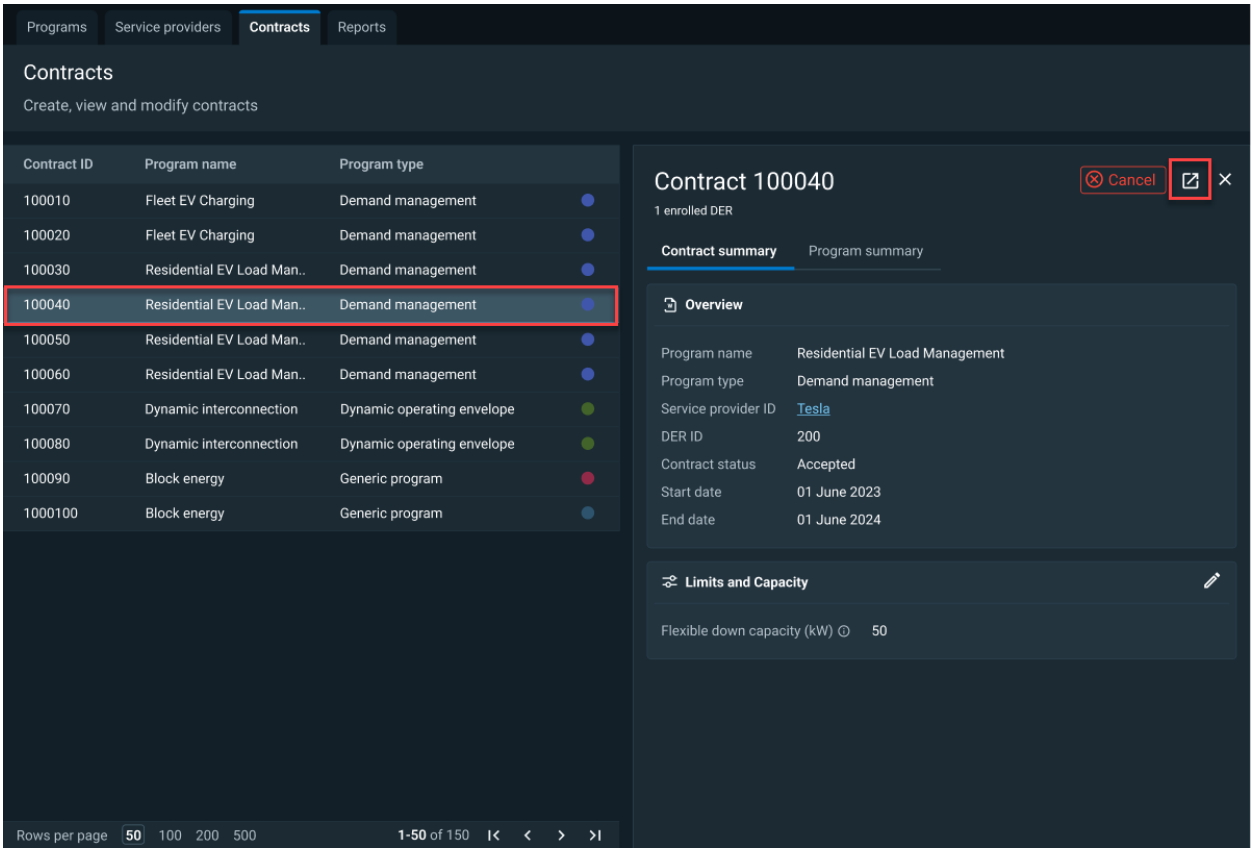
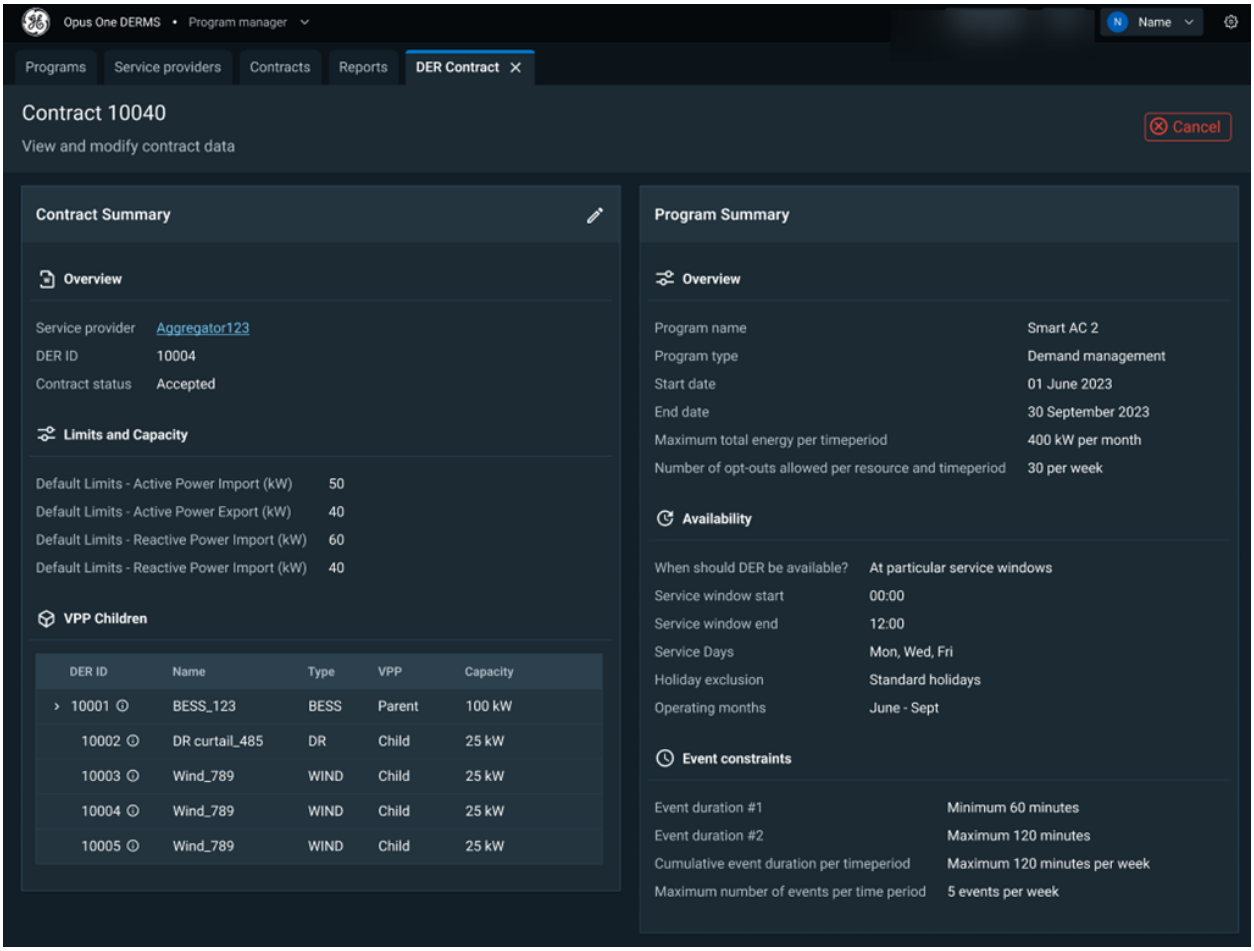


Figure 24. Contracts tab with side panel detail

Click the Expand icon to view full contract and program details in a new tab.



7.1. Cancelling a Contract

From the **Contract summary** tab on the side panel, click **Cancel** to cancel the contract. Canceling a contract unenrolls the DER from its program. You can undo a manually cancelled contract to re-instate it.

7.2. Modifying a Contract

From the **Contract summary** tab on the side panel, you can modify some limits and capacity parameters for the contract by clicking the **Edit** icon in that section of the panel. You can also modify capacity parameters of contracted DER while the contract is active.

7.2.1. Modifying Capacity Limits for Contract DER Assets

If the contract is part of a demand management program, you can modify the flexible down capacity (in kW) for the contracted DER asset. Flexible down capacity means that the contracted DER must be capable of generating the target amount of electricity during the availability timeframes in order to reduce or meet demand by that amount. In this example, the target for the contracted DER is to reduce demand by 50 kW.

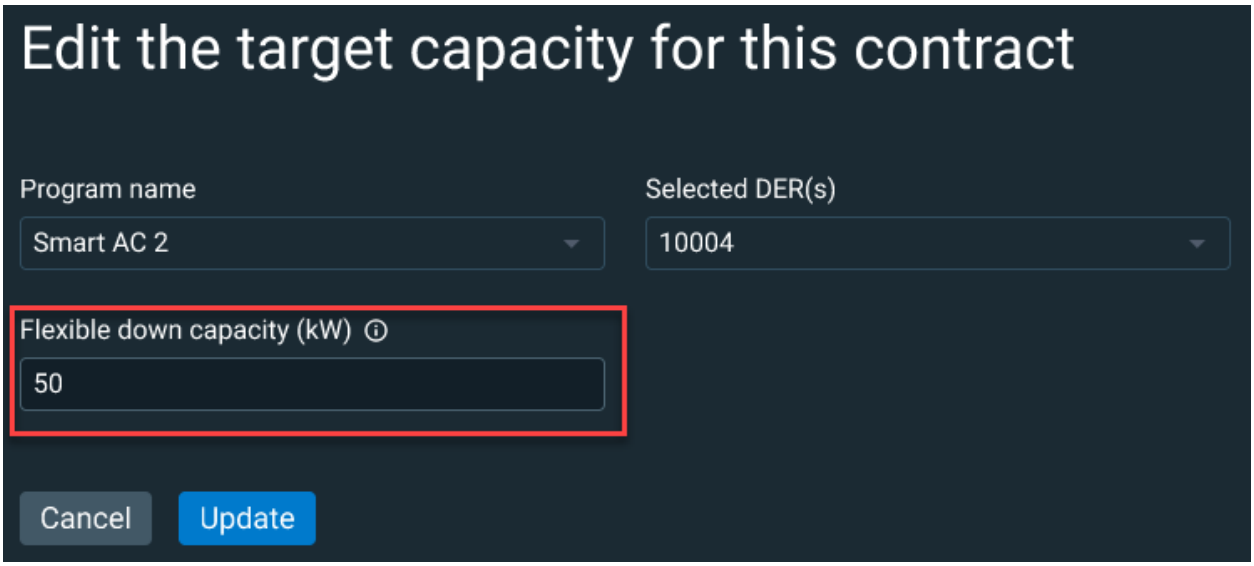


Figure 25. Defining flexible down capacity for a selected DER

To modify DER capacity in a contract:

1. Open the contract details in a new tab .
2. From the Contract Summary section of the page, click the **Edit** icon.
3. Modify the flexible down capacity amount, in kW, for the DER asset.
4. Click **Update**.

If the contract is part of a *dynamic operating envelope* program, you can edit the default capacity limits for active power and reactive power for the contracted DER asset.

To modify DER default limits in a contract:

Open the contract details in a new tab.

1. From the Contract Summary section of the page, click the **Edit** icon.
2. Modify the import and export limits for active power, in kW, for the DER asset.
3. Modify the import and export limits for reactive power, in kW, for the DER asset.
4. Click **Update**.

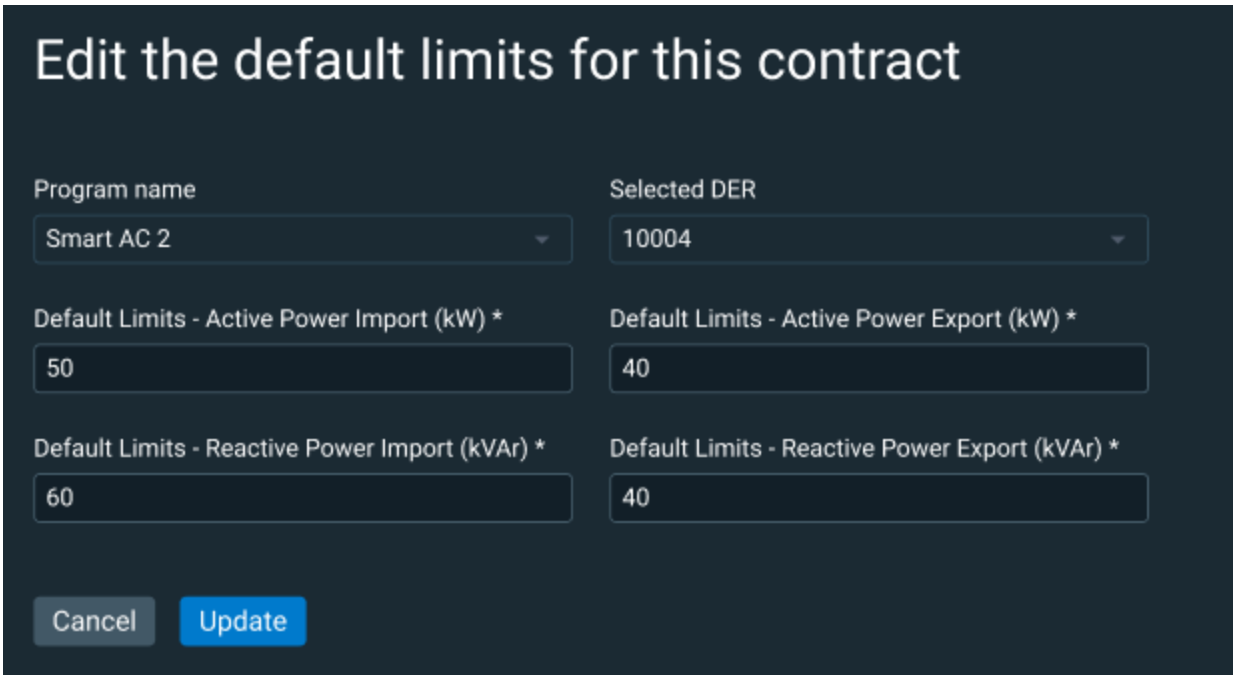


Figure 26. Editing the default limits for a contract in a dynamic operating envelope program

> **Note**
>
> There are no capacity parameters available to modify if the contract is part of a generic program.

8. Report Management

The reporting function allows you to view aggregated performance data, such as event tracking, events per program, and dispatched energy by program, for a defined time period.

On the main **Reports** tab, the top three charts—Event tracking, Events per program, and Dispatched energy—show the aggregated data of all events in the selected reporting period.

Custom-created reports are listed in table below these. Select a report from the table to view the report details in a side panel.

In addition to this aggregated data, you can also create custom reports based on individual programs or individual service providers.

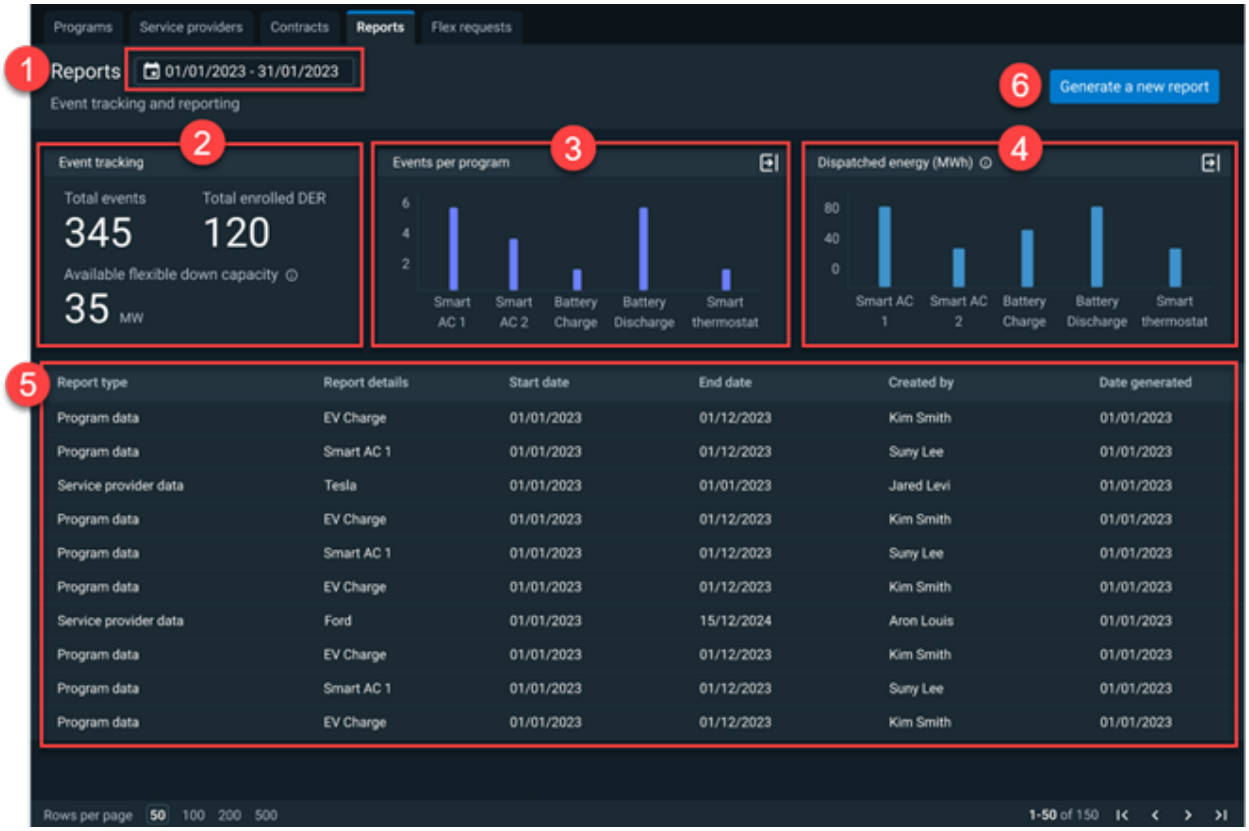


Figure 27. Reports tab

1. **Reporting time period** – Select a start and end date to define the period of data reporting you want to see.
2. **Event tracking** – Displays the count of total events, total enrolled DER, and aggregated available flexible down capacity across all programs during the reporting time period.
3. **Events per program** – Displays a bar chart with total event counts per program during the reporting period. Click the Expand/Collapse icon to see the full chart.

4. **Dispatched energy (MWh)** – Displays a bar chart with total actual energy generated (or consumed) by DER per program during the reporting time period. Click the Expand/Collapse icon in this chart to see the full chart.
5. **Custom reports** – This table lists the created reports, the reporting period, the person who created the report and when it was generated. You can keep historical reports indefinitely, or delete them.

8.1. Generating Custom Reports

You can also create reports based on individual programs or individual service providers. Reports provide information on events dispatched in Gateway and progress toward the constraints defined at program setup. Violations found against the constraints are also shown in the report. Reports help you to understand how a program is utilized, whether DERs have been performing when they are dispatched, and if the program constraints need to change for the next iteration of the program.

8.1.1. Generating a report

From the **Reports** tab, click **Generate a new report**. The **Create a new report wizard** opens.

1. Select a **Report type**. **Individual Program** or **Individual Service provider**.
   a. If the report is for a Program, select a **Program name**: A specific program to report on from the dropdown list.
   b. If the report is for a Service Provider, select a **Service provider name** from the dropdown list.
2. Select **Start** and **End** dates for the reporting period.

   > **Note**
   >
   > The **End** date cannot be in the future.

3. Click **Generate**.

For the reporting period selected, the report displays the following charts and tables.

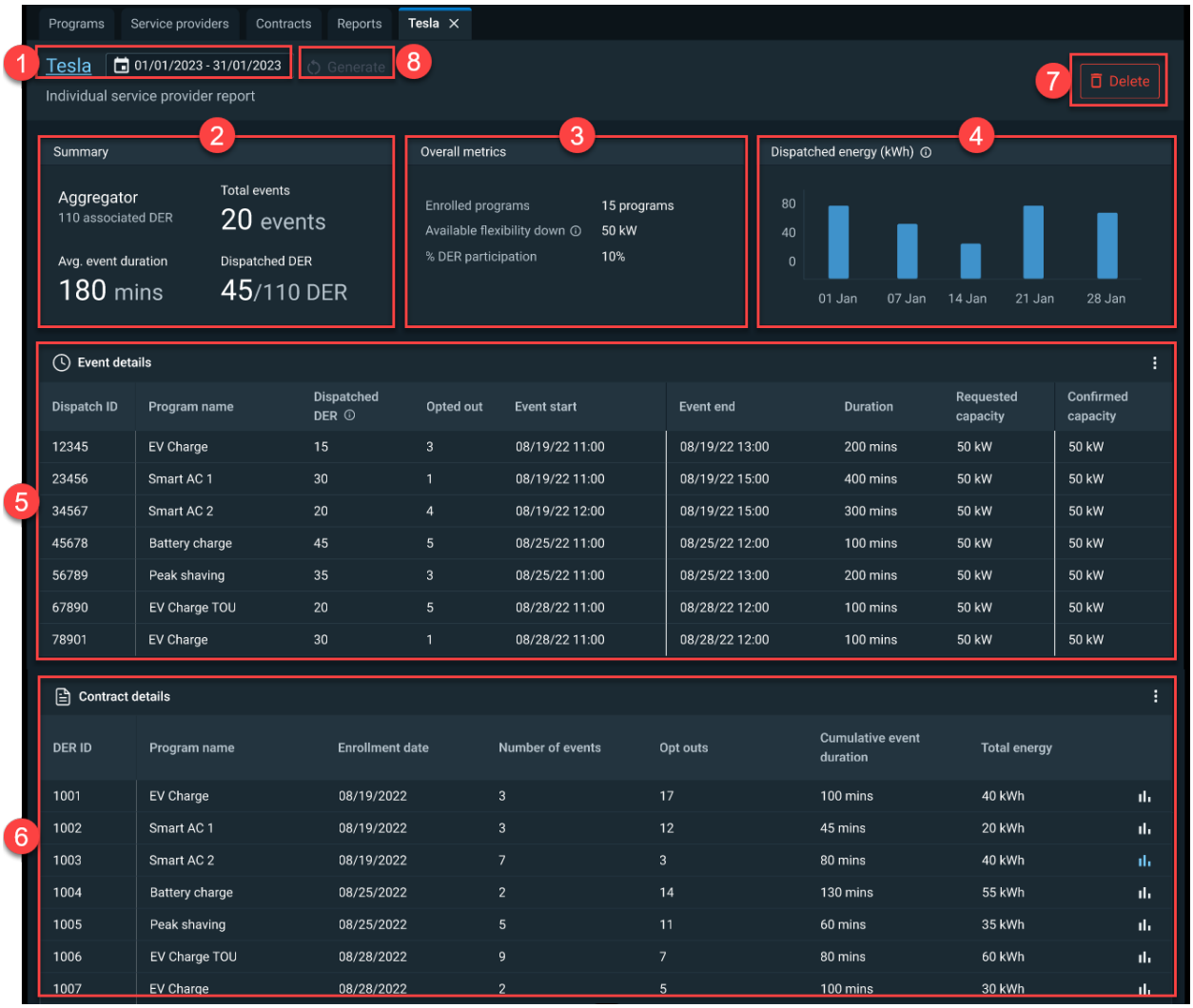


Figure 28. New custom report

1. Report name and reporting data date range, as configured.
2. A summary of high-level metrics:
   a. Aggregator: Total number of DERs associated with the select service provider during the reporting period.
   b. Total events: the total number of event dispatched during the reporting period.
   c. Avg. event duration: The average duration of all events dispatched during the reporting period.
   d. Dispatched DER: The number of DER actually dispatched out of the total number of DER associated with the service provider, based on program configuration (accounting for opt-outs and other constraints).
   e. Click on the **Dispatch ID** hyperlink, side panel displays. You can view VPP details in the side panel.

Event details

Dispatch ID	Event start	Event end	Duration	Dispatched DER	Opted out DER	Requested capacity	Confirmed capacity
12345	08/19/22 11:00	08/19/22 13:00	200 mins	15 DER	2 DER	50 kW	50 kW
23456	08/19/22 11:00	08/19/22 15:00	400 mins	23 DER	3 DER	50 kW	50 kW
34567	08/19/22 12:00	08/19/22 15:00	300 mins	23 DER	1 DER	50 kW	50 kW
45678	08/25/22 11:00	08/25/22 12:00	100 mins	5 DER	2 DER	50 kW	50 kW

Figure 29. Dispatch ID hyperlink

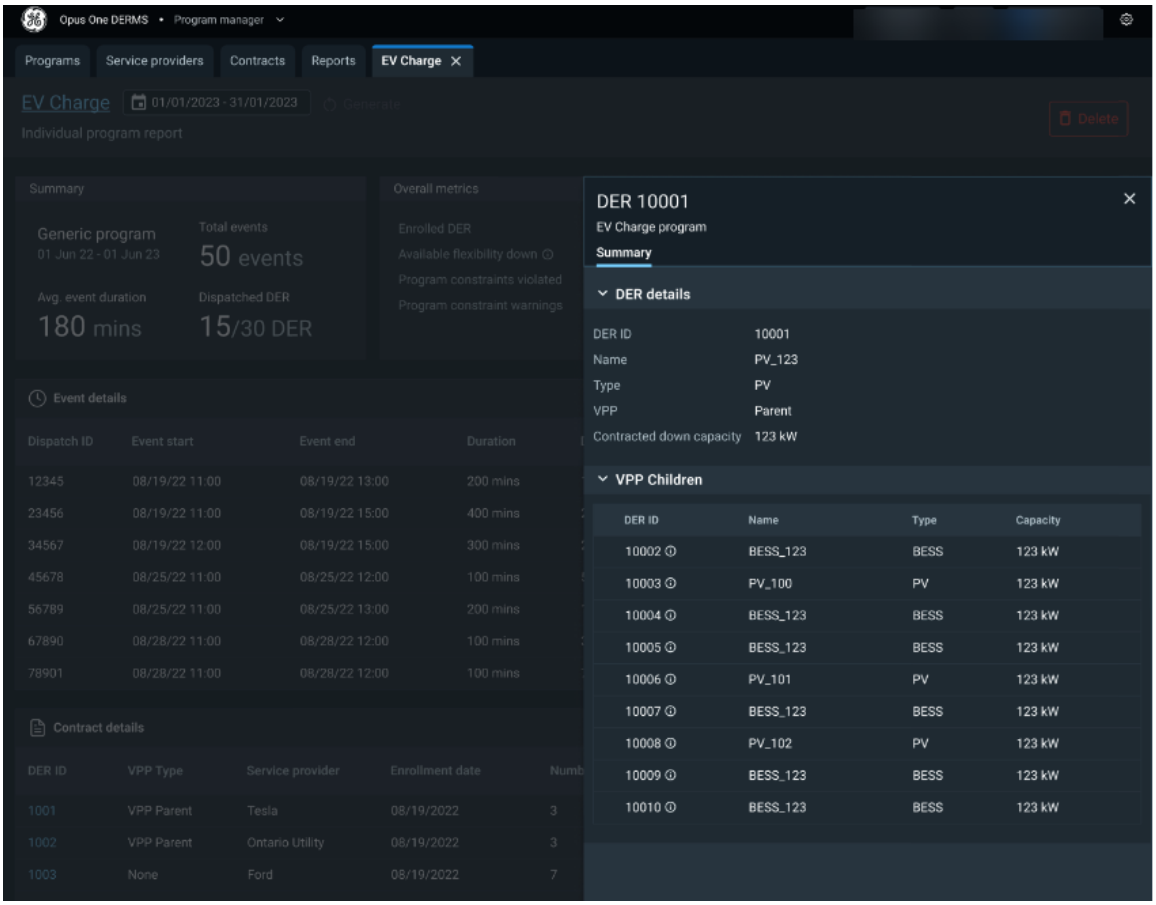


Figure 30. Dispatch ID Side Panel

3. Overall metrics: Overall service provider metrics: Actual number of programs enrolled, total available flexible down capacity, and overall percent DER participation in a program during the reporting period.
4. Dispatched energy in kWh over the reporting period.
5. Event details: A table of event metrics on a per-dispatch basis, with confirmed dispatched capacity.
6. Contract details: Contracted event details on a per-DER basis.
7. Delete report.
8. Generate: Generate a new report with the defined report criteria.

9. Virtual Power Plant (VPP)

The Model Manager database supports modeling of the parent-to-child relationship between the VPP DER (“parent”) and its constituent DERs (“children”). Measurement and status data is collected for constituent DERs as it is for any other DERs and sent to the DMS so their impact to load flows can be correctly accounted for. Aggregate measurements and status for VPP DERs are provided by its Aggregator. Only the aggregate DER (that is, the VPP “parent”) can be enrolled in programs and be made available for dispatch.

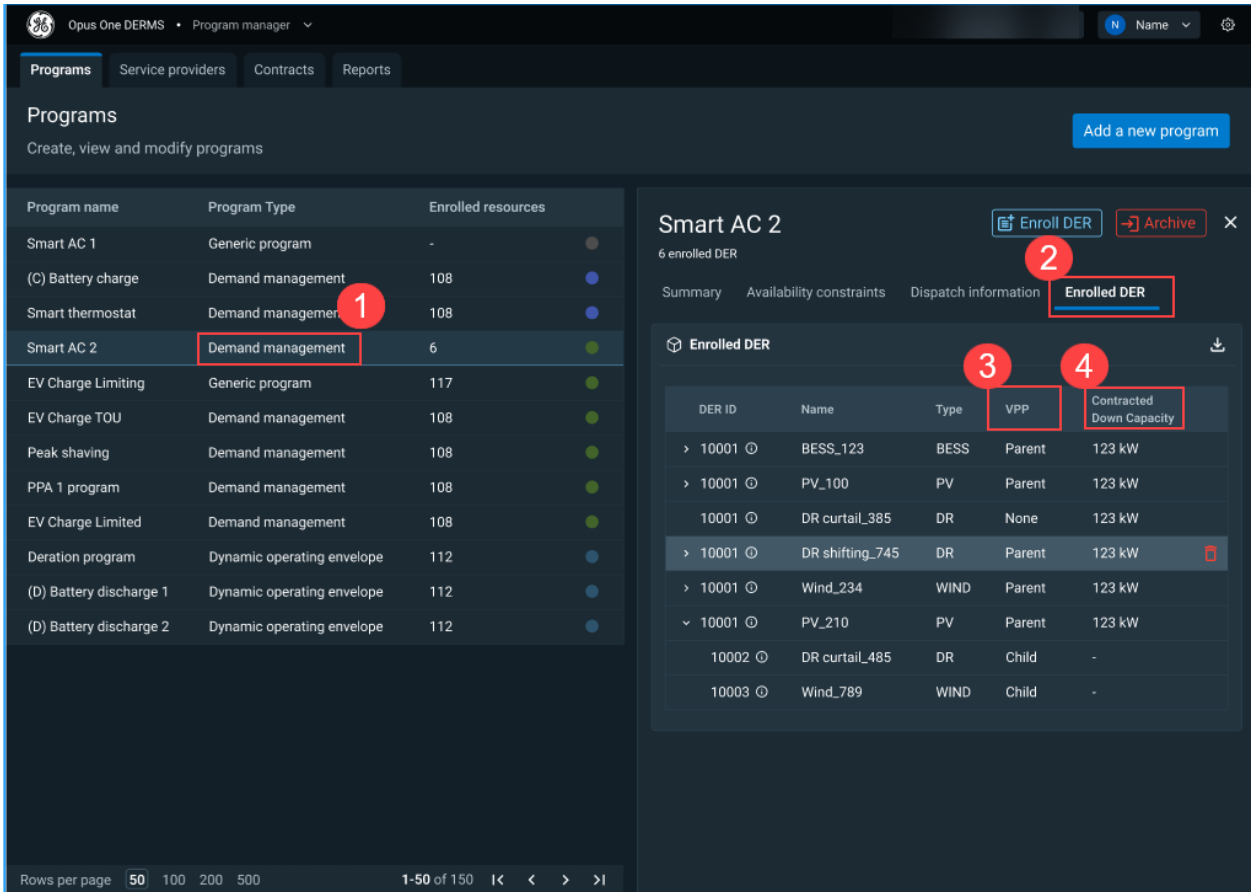
In Program Manager, you can now view the VPP details here in multiple features of the application

9.1. Programs

In the Programs tab, to view VPP details:

1. Select any program In the side panel window, click the Enrolled DER tab.
2. VPP and Contracted Down Capacity columns are displayed under the Enrolled DER details.

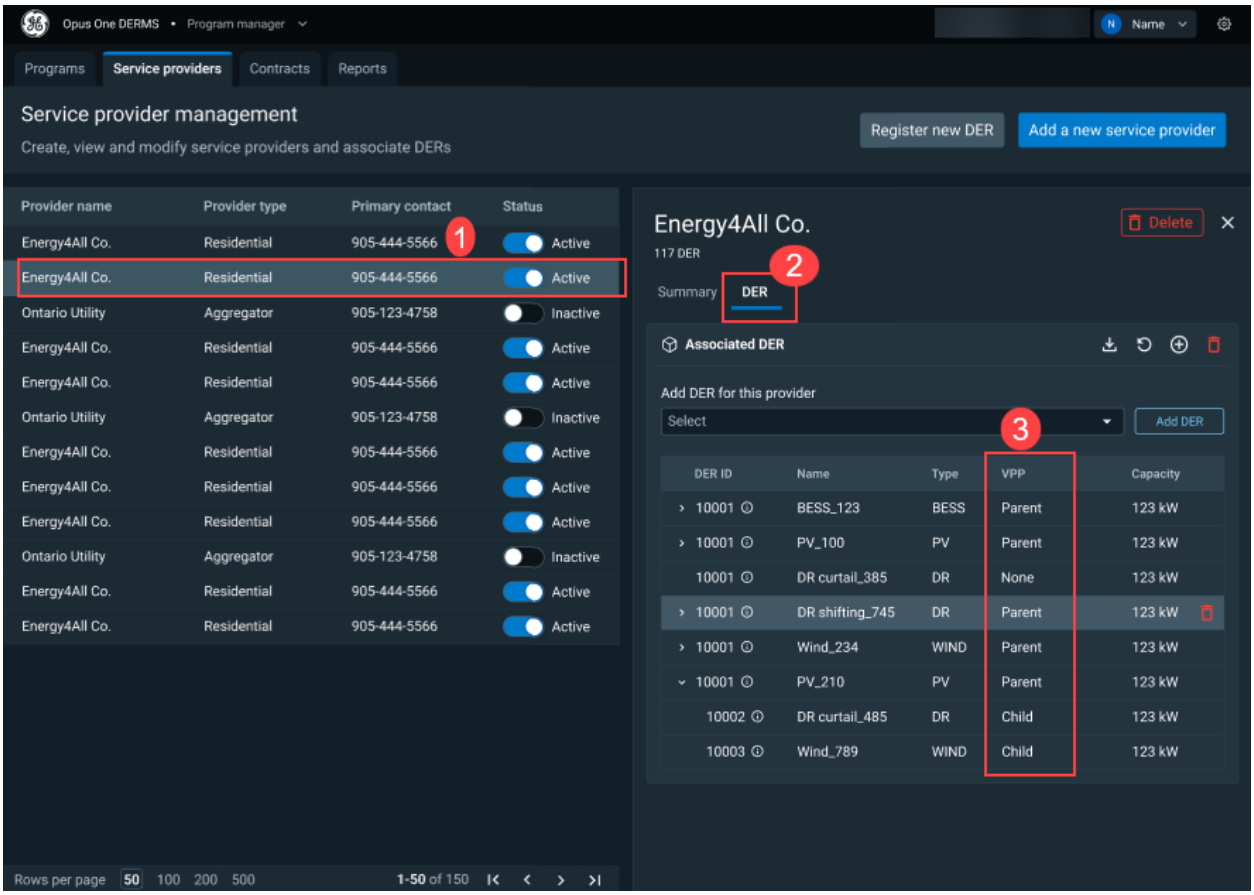
> **Note**
>
> If the Program Type is not Demand Management, the Contracted Down Capacity column is not displayed in the Enrolled DER details window.



9.2. Service Provider

To view VPP data:

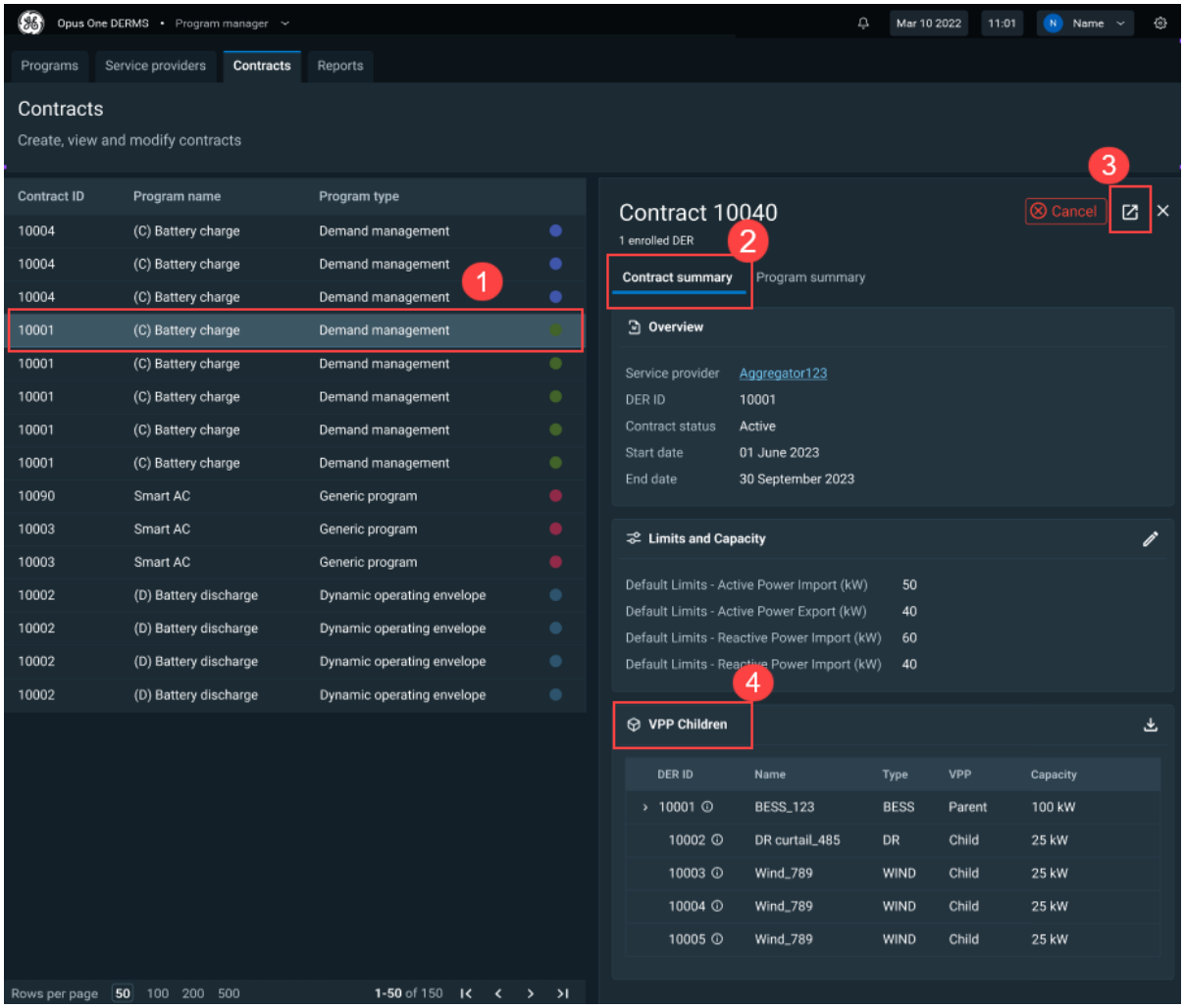
1. Select a listed program in the tabular view.
2. In the side panel window, click the DER tab.
3. VPP column with the defined parent and child information is displayed.



9.3. Contracts

You can track the accurate enrollment of DERs and VPPs. All the changes made in the DER Warehouse (i.e.) change a VPP child or parent, all these changes are reflected in the program data. To view the data:

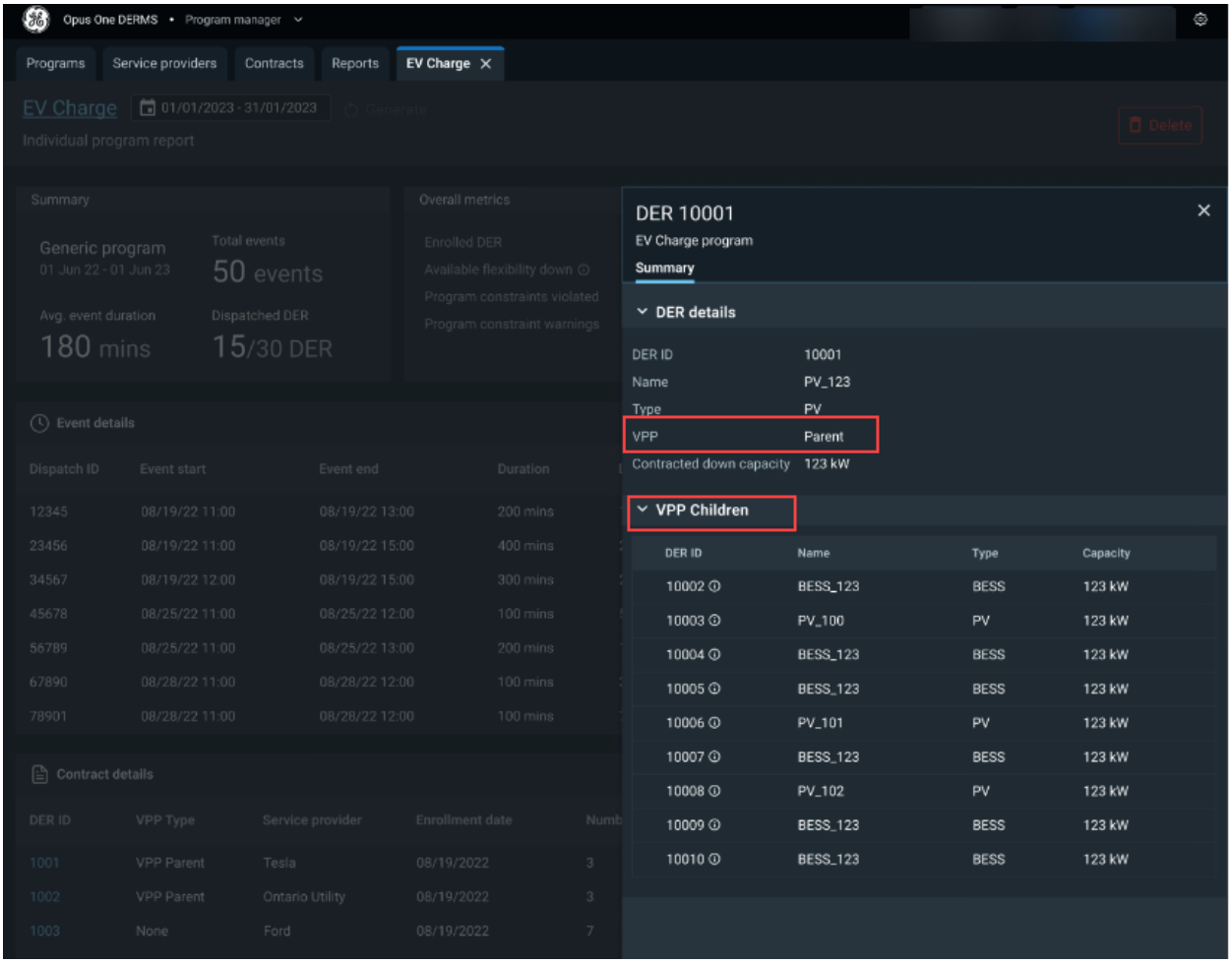
1. Select a listed contract in the tabular view.
2. In the displayed side panel window, click Contract Summary tab to view VPP data.



3. Click the Expand/Collapse icon to see the full chart as shown in the below image.

9.4. Reports

In the Reports tab, run an event for the program, to view the event details including the VPP parent and the children that were dispatched.


Opus One DERMS • Program manager
Programs
Service providers
Contracts
Reports
EV Charge
EV Charge
01/01/2023 - 31/01/2023
Generate
Delete
Individual program report
Summary
Generic program
01 Jun 22 - 01 Jun 23
Total events
50 events
Avg. event duration
180 mins
Dispatched DER
15/30 DER
Overall metrics
Enrolled DER
Available flexibility down
Program constraints violated
Program constraint warnings
Event details
Dispatch ID
Event start
Event end
Duration
12345 08/19/22 11:00 08/19/22 13:00 200 mins
23456 08/19/22 11:00 08/19/22 15:00 400 mins
34567 08/19/22 12:00 08/19/22 15:00 300 mins
45678 08/25/22 11:00 08/25/22 12:00 100 mins
56789 08/25/22 11:00 08/25/22 13:00 200 mins
67890 08/28/22 11:00 08/28/22 12:00 100 mins
78901 08/28/22 11:00 08/28/22 12:00 100 mins
Contract details
DER ID
VPP Type
Service provider
Enrollment date
1001 VPP Parent Tesla 08/19/2022 3
1002 VPP Parent Ontario Utility 08/19/2022 3
1003 None Ford 08/19/2022 7
DER 10001
EV Charge program
Summary
DER details
DER ID 10001
Name PV_123
Type PV
VPP Parent
Contracted down capacity 123 kW
VPP Children
DER ID
Name
Type
Capacity
10002 BESS_123 BESS 123 kW
10003 PV_100 PV 123 kW
10004 BESS_123 BESS 123 kW
10005 BESS_123 BESS 123 kW
10006 PV_101 PV 123 kW
10007 BESS_123 BESS 123 kW
10008 PV_102 PV 123 kW
10009 BESS_123 BESS 123 kW
10010 BESS_123 BESS 123 kW